When climate and institutions narrow the feasible set

Three rigorous economics papers · two applied development studies and one classic institutions read. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

Across all three papers, shocks and rules matter because they change the **set of choices people can actually afford**. In rural India, warming reduces agricultural resources and appears to constrict marriage-related migration, particularly where dowry norms are strongest. In Rwanda, weather-suited malaria exposure depresses high-stakes test performance and widens a gender gap, while bed nets soften the loss. Bruhn's Mexican classic supplies the institutional counterpoint: cutting registration costs creates firms and jobs, but mainly by inducing wage earners to become entrepreneurs rather than by formalizing incumbents. The literature is increasingly strongest when it links a credible source of variation to a specific household or firm margin, then tests the mechanism instead of stopping at an average effect.

A district panel combining three Indian censuses, household surveys, and local weather asks whether long-run warming changes women's marriage-related migration. District and year fixed-effects estimates link higher temperatures to less rural-rural and rural-urban female migration, especially in high-dowry areas; lower crop yields, consumption, and dowry resources are plausible channels. The paper's gendered climate mechanism is compelling, though the identifying variation is observational and remaining district-specific trends or migration measurement limits warrant caution.

More than one million Rwandan primary-school exam records are matched to a weather-driven malaria-suitability index. School and year fixed effects, rich weather controls, timing tests, spatial-error checks, trimming exercises, and disease-placebo tests support a sizable negative relationship with test scores; girls are more affected, while bed-net coverage attenuates losses. Weather suitability is an ingenious exposure measure, but it is still a reduced-form design and may not isolate every channel through which temperature affects learning.

Mexico's rapid-business-opening system reached municipalities at different dates, creating a staggered difference-in-differences design with quarterly employment microdata. Registration rises about 5 percent in eligible industries and employment about 2.8 percent, driven by former wage earners starting firms rather than existing informal owners formalizing; new competition also lowers prices and incumbent income. Adoption timing is the key identifying assumption, and the observed post-reform horizon is short, but the channel evidence makes this a durable applied-economics classic.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Scorching Heat and Shrinking Horizons: The Impact of Rising Temperatures on Marriages and Migration in Rural India

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Abstract

This study delves into the gendered impacts of rising temperatures from climate change on long-term female migration patterns in rural India. We utilize a panel fixed-effects model to investigate the relationship between temperature levels and migration trends by employing a district-level panel dataset that combines Census data for years 1991, 2001, and 2011, multiple rounds of household surveys, and local weather measurements. Our results showcase statistically significant declining effects of rising temperatures on female rural-rural and rural-urban migration in the country. Specifically, the findings suggest that a 1 °C temperature increase is associated with a 22% decline in rural-urban and a 13% decline in rural-rural female migration in an average district in India. This decline is primarily due to diminishing marriage-related migration among women in the northern districts of the country, which have a historically high prevalence of dowry. We identify decreasing agricultural yields from rising temperatures as an underlying mechanism that is reducing resources to finance dowry in northern India. We further note that the declines in female migration are driven by districts with poor access to credit, and more so in the northern districts.

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I Introduction

Anthropogenic climate change is one of the most pressing issues of our time, with far-reaching impacts on human society. According to the sixth assessment report of the Intergovernmental Panel on Climate Change (IPCC), global warming of 1.1 degrees Celsius above pre-industrial levels is already profoundly affecting human systems (IPCC, 2021). These effects are felt most acutely in underdeveloped countries located in low-latitude regions and near the equator (Castells-Quintana et al., 2017; Cline, 2007; Dell et al., 2012). In this regard, Carleton and Hsiang (2016) argue that climate change, by variably affecting different sub-populations like women and the poor, can lead to substantial changes in demographic structures in the long run. These effects can potentially result in further distortion of existing social and economic inequalities in developing countries.

One such demographic effect of climate change is on migration patterns, which has garnered significant attention in recent studies (Adger et al., 2015; Black et al., 2011; Cattaneo et al., 2019; Hoffmann et al., 2021). However, while the literature has extensively discussed the impacts of climate change on migration rates in the global south, there continues to be limited evidence on the gendered impacts on migration. This is critical, given the prevalence of various gender-related norms surrounding women's migration in developing countries in South Asia and Africa (Bau, 2021; Khalil & Mookerjee, 2018). In light of this, in this study, we investigate the long-run impacts of climate change on internal female migration in India and explore the mediating role of marriage in shaping those impacts. To do this, we create a unique database combining information on intra-district rural-urban and rural-rural female migration for the years 1991, 2001, and 2011 from the Census of India and district-level data on temperature and precipitation from ERA5 reanalysis weather data (Berrisford et al., 2011). Additionally, we gather multiple waves of household-level surveys by the National Sample Survey (NSS) of India between 1987 and 2011 to further delve into the dynamics.

The gendered effects of climate change on migration are quite significant. We find that a 1°C increase in mean decadal temperature is associated with a 22% decline in female rural-urban migration and a 13% decline in female rural-rural migration in an average Indian district. Our results are also robust to multiple sensitivity tests. They remain robust to the inclusion of state-year trends and when the spatial correlation between the districts is taken into account using Conley standard errors (Conley, 1999). The findings are critical as India has a patrilocal structure where most women migrate to their in-laws' houses after marriage (Chatterjee & Desai, 2020). In fact, most migrants in India are women who migrate because of marriage and not for economic reasons, which is also due to the low participation of women in labor markets (Afridi et al., 2018; Mehrotra & Parida, 2017).

Moreover, the event of marriage in India includes the traditional custom of dowry, in which the bride's household transfers wealth and resources to the groom's household during marriage. Despite legal procedures, the custom continues to be highly pervasive in the country(Chiplunkar & Weaver, 2023) and plays a decisive role in a woman's marriage. In this study, we showcase that the declining effects on female migration are driven by reductions in female marriage migration, primarily in the northern part of the country where the dowry custom has been historically more rooted (Mitchell & Soni, 2021). We find that falling agricultural yields due to rising temperatures (Taraz, 2018) that are eroding agricultural incomes act as an underlying mechanism driving our results. Moreover, we also document that poor access to credit in rural areas is moderating the impacts of increasing temperatures on female migration. Besides, although we can only examine impacts on intra-district female migration due to data limitations, multiple sets of evidence indicate that most marriages occurred within the short distance of a woman's native place and within the same district during the time period of our study (Chiplunkar & Weaver, 2023; Fulford, 2013)[1], which upholds the findings of our study. The results of our study highlight a concerning trend as for many women in India, moving to the city through marriage is a means to broaden their horizons and improve their living standards (Kalpagam, 2008). Climate change, by interacting with traditional gender norm, is insidiously shrinking the prospects of women's movement to urban areas through marriage, trapping women in rural regions with potential negative consequences on their welfare.

Our study makes a significant contribution to multiple strands of literature. We supplement the literature on the demographic effects of climate change (Carleton & Hsiang, 2016). We provide evidence of how climate change is distorting demographic structure in the long run by disproportionately affecting the female sub-population in rural India. We also advance the growing literature on the effects of climate change on migration patterns in developing countries (Cattaneo et al., 2019; Hoffmann et al., 2021). Specifically, our study is related to works by Gray and Mueller (2012) and(Findley, 1994). Gray and Mueller (2012) examine the consequences of droughts on migration in rural Ethiopia, and note that drought events are associated with declines in female out-migration as households have fewer resources to bear wedding expenses as traditional norms dictate that the bride's household are responsible for spending on wedding expenses. Findley (1994) documents similar observations for Mali, where the occurrence of droughts is related to decreases in migration by women as resources to arrange weddings shrink. In our study, we show that rising temperature levels are associated with falling female marriage-related marriage migration in rural India. In this way, we also add to the literature on the broader economic implications of the increasing temperature levels in rural India. These studies include shrinking rural consumption levels (Aggarwal, 2020; Liu et al., 2023; Sedova et al., 2020), rising rural mortality rates (Burgess et al., 2014), and intensification of agricultural activities as demand for non-agricultural products shrivels due to falling agricultural incomes (Liu et al., 2023).

Moreover, our study augments the literature on how traditional gender-related norms shape gender outcomes in developing countries. For instance, Ashraf et al. (2019) document that in Indonesia and Zambia, groups that practice the custom of bride price tend to have more educated daughters. Lowes and Nunn (2017) note no associations between larger bride prices and earlier marriages and high fertility but even assert that bride price is linked to better quality marriages and high self-reported happiness from the wife. In contrast, in India, studies have recorded perverse gender outcomes due to dowry. For example, Bhalotra et al. (2020) detect that higher dowry costs, reflected in the high cost of gold, result in increased son preference behavior amongst the parents. Sekhri and Storeygard (2014) observe that dry rainfall shocks are linked to increases in the incidence of domestic violence against women and dowry deaths in India[2]. This is because dowry payments are used to smooth consumption in the event of dry shocks by the groom's households. Moreover, Calvi and Keskar (2021) conclude that women

[1] This is discussed further with evidence in Section II.II.

[2] According to Sekhri and Storeygard (2014) , in India, a dowry death is legally defined as the death of a woman that occurs within seven years of her marriage and is caused by any burns or other bodily injury that do not occur under normal circumstances.

and resorting to paying higher dowries to secure them are documented in earlier literature as well, such as in Caldwell et al. (1983) . Botticini and Siow (2003) quote Caldwell's observation as follows:

> Parents desire their daughters to marry educated men with urban jobs because such men have higher and more certain incomes, which are not subject to climatic cycles and which are paid monthly, and because the wives of such men will be freed from the drudgery of rural work and will usually live apart from their parents-in-law. In a sellers' market created by relative scarcity, there was no alternative but to offer a dowry with one's daughter (Caldwell et al., 1983, p. 347).

The practice of paying higher amounts of dowry to secure urban grooms has become common, and without sufficient dowry, women may have to settle for "low quality"grooms. In their study, Kalpagam (2008) describes a situation where a family's financial inability to provide a substantial dowry resulted in their daughter marrying a casual laborer in the village, who frequently faced unemployment, underlining the significant impact dowry can have on the marital prospects of women in rural regions. Besides, the preference for urban grooms not only stems from rural parents but also from the daughters, who associate urban life with modernity and freedom to lead their private lives, experience unrestricted mobility, freedom in style and clothing, and generally better quality of life. These differences in the rural and urban life of a woman are reflected in statistics based on the India Human Development Survey (IHDS) presented in Table A2 in the Appendix. Most women in both urban and rural regions need permission from their husbands to go outside, but those in urban areas are more likely to go alone once permission is granted, indicating greater freedom in mobility. Moreover, the practice of ghoonghat/purdah/veiling[3] is much more pervasive in rural areas. We also note that the norm of men eating before women is much more common in rural areas in northern India[4], which, as Sedlander et al. (2021) point out, leads to inequitable food allocation between men and women in the household.

These discussions highlight that rural parents and daughters prefer urban grooms, employed in the non-agricultural sector, for which parents agree to pay high amounts of dowry. As explained in subsequent sections, the increasing temperatures due to climate change, which is harming agricultural yields and incomes, can severely impact the wealth needed to finance these dowries. Coupled with limited access to credit in rural areas, increasing temperatures can further curb resources to finance dowry. In the long run, this can harm the prospects of rural women getting married to urban grooms, leading to significant changes in female marriage migration patterns in India.

III Data Sources and Methodology

This section details the main data sources employed in our study and describes the empirical strategy implemented to examine the effects of rising temperatures on female migration rates.

[3]Ghoonghat or purdah refers to the practice of covering the head and/or face amongst married women to seclude their contact with men outside the family and sometimes inside the family such as with relatives (Desai & Andrist, 2010).

[4]This is also highlighted by Sedlander et al. (2021) , who discuss how this traditional norm of men eating before women in rural India is linked to inequitable food allocation and lower health outcomes for women in the country.

III.I Data Sources

We use three main data sources in the study and assemble a district-level panel dataset by combining data on migration rates and weather variables. First, we formulate district-level measures of female rural-rural and rural-urban migration rates using the Indian Census data. Second, we aggregate the gridded monthly weather data published by the European Centre for Medium Range Weather Forecasts (ECMWF) (Berrisford et al., 2011) to construct district-level measures of temperature and precipitation. Third, we utilize micro-level household survey data from the National Sample Surveys (NSS) of India, conducted between 1987 and 2012. Using NSS, we extract district-level information on marriage migration by women. Together, these data sources allow us to construct a rich dataset to examine the impact of rising temperatures on female migration patterns in rural India and to test the effects on marriage-related migration.

III.I.1 Data on climate

For the climate data, we employ gridded monthly data on temperature and precipitation from the ERA5 high-quality reanalysis data produced by the European Centre for Medium Range Weather Forecasts (ECMWF) (Berrisford et al., 2011). Following the recent climate literature, we compute the average value of all grid points falling within a district boundary. Besides, as a robustness test in the later sections (Section IV.I), we also use an alternative aggregation method in which the grid-level data is aggregated to district-level weather outcomes, that is, mean temperature and mean precipitation by calculating the weighted average of all grid points within 100 kilometers of each district's geographic center using inverse-square weighting (Taraz, 2018).

The main variables are annual average temperature and average precipitation at the district level, and these variables are computed for each year between 1981 and 2010. The spatial distribution of the changes in the temperature and precipitation between the years 1981 and 2010 is presented in Figure 1. It is clear that almost all districts have experienced a rise in average temperature levels by at least 0.1 °C during the period of analysis. However, the figure shows a substantial spatial difference in temperature increases that can be utilized for our study. The descriptive statistics of these variables are presented in Table A1 in Appendix A.

III.I.2 Data on female rural-rural and rural-urban migration

To construct the main dependent variables capturing the female rural out-migration rates, we use the Census of India data for the years 1991, 2001, and 2011. We gather the district-wise population data using the Primary Census Abstract (PCA) data tables. The data on migration are compiled from the Migration (D-series) data tables from the Census digital library[5]. These tables provide district-level data on the total female population and female intra-district rural-urban and rural-rural migrants. For completeness and comparison, we have also collected information on the total and male population in the district and total and male intra-district rural-urban and rural-rural migrants.

To ensure consistency in district boundaries across the years, we harmonize the district boundaries using the district mapping file provided by Kumar and Somanathan (2017) . There were 466 districts in India in 1991. Figure A.1 in the Appendix depicts these 466 district boundaries that are consistent between 1991, 2001, and 2011. We could not include 14 districts from the state of Jammu and Kashmir[6] in the analysis due to the unavailability of census data in 1991. Additionally, due to the unavailability of weather data, we dropped 4 districts of union territory Pudducherry and 1 district of Lakshadweep, leading to a final sample of 447 districts.

The primary dependent variable is defined as the share of rural-urban (rural-rural) female intra-district migrants in the total female population of a district. Specifically, it is defined as:

$$Y_{j,t} = \frac{M_{j,t}}{P_{j,t}} \tag{1}$$

where $M_{j,t}$ is the number of rural-urban (rural-rural) female migrants and $P_{j,t}$ is the total female population in district j and in the decade beginning with year $t(= 1991, 2001, 2011)$. We construct this variable for female intra-district rural-urban (rural-rural) migrants as it is our primary outcome of interest. For comparison, we also construct similar variables measuring involving total and male intra-district rural-urban and rural-rural migrants.

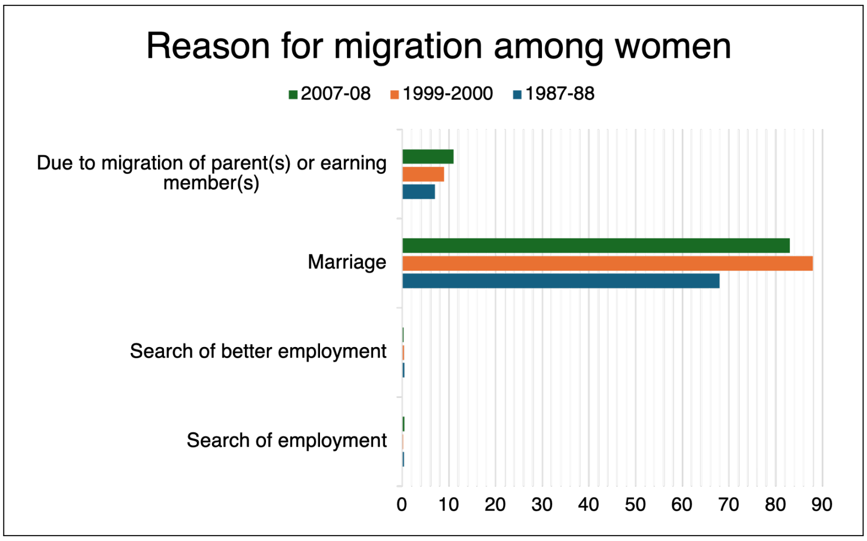
We present these variables in Figure 2, which highlights two important aspects of internal migration in India: (1) Intra-district rural-rural migration by both men and women is more common than rural-urban migration; and, as previously discussed (2) Women make up the majority of internal migrants from rural areas in the country, irrespective of whether they move to other rural or urban destinations.

[5] https://censusindia.gov.in

[6] The state now is organized into two union territories: Jammu and Kashmir and Ladakh

NSS-EUS provides additional evidence that most internal migration by working-age[8] married women are due to marriage. As shown in Figure 3, 68% of internal married working-age female migrants in India moved due to marriage in the 1987-88 round, which increased to over 80% in subsequent rounds. Additional descriptive statistics derived from NSS data are displayed in Figure A.3 in Appendix A.

Figure 3: Female Migration by Reason in India, NSS, 1987-2008



Note: The chart depicts the primary reason for migration among married working-age women in three NSS rounds: 1987-88, 1999-2000, and 2007-08. Source: Authors' own illustration using NSS-EUS data.

We establish a linkage between the NSS and Census data by utilizing the district codes available in the NSS surveys. To ensure that the districts are consistent with the Census data and to maintain the same district boundaries, we refer to Murthi et al. (2001) and Kumar and Somanathan (2017) .

III.I.3 Data on additional variables

The analysis is supplemented with data from several other sources. As we argue that reductions in agricultural yields due to rising temperatures act as an underlying mechanism explaining our results, we gather data on agricultural yields to test this mechanism. We collect agricultural data from the Village Dynamics in South Asia (VDSA), provided by the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT, 2015). This data set has annual measures of district-level agricultural yields for 15 crops between the years 1966 and 2010. These crops are rice, wheat, sugarcane, cotton, groundnut, sorghum, maize, pearl millet, finger millet, barley, chickpeas, pigeon pea, sesame, rapeseed and mustard, castor, and linseed. Following Taraz (2018) , we derive a price-weighted composite yield measure at the district level for each year

of women has fallen in the same time frame. Rao and Finnoff (2015) have conclusively shown that these trends are indeed due to rises in marriage migration and not increases in migration by women for economic purposes that are disguised as marriage migration. Mazumdar et al. (2013) also discuss this issue and using primary survey data and field studies conclude that labor migration by women in India is very low and most migration by women is for marriage.

[8]Following Rao and Finnoff (2015) , we consider working-age to be between 15 and 64 years old.

climate change literature on India, which documents consistent rises in average temperatures across the country but region-specific changes in precipitation, we only interpret the coefficients associated with the temperature levels throughout this study. Thus, β_0 is the main coefficient of interest in this study and captures the effect of rising temperatures on migration rates, and $\beta_0 < 0$ signifies that migration rates are declining with rising temperatures.

Furthermore, we also include fixed effects, ϕ_j, capturing fixed district characteristics. γ_t is the year effects that control for changes over time that are consistent across districts. We only include fixed effects as controls to ensure that the model stays parsimonious and to avoid overcontrolling. This is because including other controls, such as socioeconomic characteristics that might be correlated to agricultural productivity, may introduce bias in the estimation due to over-controlling (Cattaneo & Peri, 2016). The standard errors are clustered at the district level and are provided in parentheses in the regression tables.

Specification (2) is a reduced-form linear relationship between rural-urban and rural-rural female migration rates and climate variables. In the upcoming sections, we discuss the possible channels through which climate change may shape migration rates, such as changes in agricultural productivity and contextual factors like access to credit.

IV Results

Table 1 illustrates the estimated effects of annual temperatures on intra-district female rural-rural and rural-urban migration rates in India. They indicate a statistically significant negative effect of increasing temperatures on rural-rural and rural-urban female migration rates. Specifically, the results show that a 1 °Celsius increase in temperature in an average Indian district is associated with a 22% decrease in the share of female rural-urban migrants and a 13% reduction in the share of female rural-rural migrants. The effects on male migration rates are not statistically significant, aligning with the findings of Liu et al. (2023) .

Table 1: Effect of Rising Temperatures on Migration, 1991-2011

	Rural-Urban			Rural-Rural		
	Total	Male	Female	Total	Male	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Temperature	−0.004	−0.002	−0.007	−0.018	−0.006	−0.030
	(0.003)	(0.003)	(0.003)**	(0.008)**	(0.006)	(0.012)**
Precipitation	0.004	0.003	0.004	−0.001	0.008	−0.010
	(0.003)	(0.002)	(0.003)	(0.005)	(0.003)**	(0.007)
District FE	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y
Observations	1,341	1,341	1,341	1,341	1,341	1,341

Note: The dependent variable in (1) is the share of total intra-district rural-urban migrants, in (2) is the share of male intra-district rural-urban migrants, in (3) is the share of female intra-district rural-urban migrants, in (4) is the share of total intra-district rural-rural migrants, in (5) is the share of male intra-district rural-rural migrants, and in (6) is the share of female intra-district rural-rural migrations, in years 1991, 2001, and 2011. The independent variables are the decadal averages of annual temperature (°C) and annual precipitation (mm). The standard errors are clustered by district in parentheses.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

We argue that the declines in female migration are primarily due to the decreases in marriage-related migration. Furthermore, these declines could be driven by the districts in northern India, where the custom of dowry, as previously discussed, has been historically widespread. To check for this, we categorize the districts as northern districts that are located in the northern states[11]. We exploit this heterogeneity in the historical prevalence of dowry between the northern districts and the rest of India by re-running the previous analysis and allowing the coefficients of temperature and precipitation to vary depending on whether the district is in a northern state. The results are reported in Table 2 and match our expectations. In the northern districts, rising temperatures have a pronounced negative effect on female migration rates[12].

Table 2: Effect of Rising Temperatures on Female Migration in Northern Districts, 1991-2011

	Rural-Urban	Rural-Rural
	Female	Female
	(1)	(2)
Temperature	0.010	0.029
	$(0.005)^{**}$	$(0.016)^{*}$
Temperature x North	-0.022	-0.075
	$(0.004)^{***}$	$(0.012)^{***}$
Precipitation	0.006	-0.007
	$(0.003)^{*}$	(0.006)
Precipitation x North	-0.017	-0.040
	$(0.004)^{***}$	$(0.015)^{***}$
District FE	Y	Y
Year FE	Y	Y
Observations	$1,341$	$1,341$

Note: The dependent variable in (1) is the share of female intra-district rural-urban migrants in the years 1991, 2001, and 2011. The dependent variable in (2) is the share of female intra-district rural-rural migrants in the years 1991, 2001, and 2011. The independent variables in (1) and (2) are the decadal averages of annual temperature (°C) and decadal averages of annual temperature interacted with "North" dummy denoting 1 if the district lies in a major northern state such as Punjab, Uttarakhand, Uttar Pradesh, Haryana, Bihar, Jharkhand, Rajasthan, Madhya Pradesh, and Chhattisgarh, and 0 otherwise. It also includes the decadal averages of annual precipitation (in mm) and decadal averages of annual precipitation interacted with the "North" dummy. The standard errors are clustered by district in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

It might still be the case that these changes observed through Census data are not due to

[11]We follow the Government of India's categorization of the north and north-central cultural zone (Ministry of Culture, Govt. of India, n.d.) to identify the northern states and classify the states Punjab, Haryana, Uttar Pradesh, Rajasthan, Bihar, Rajasthan, Uttarakhand, Chhattisgarh, Jharkhand, and Madhya Pradesh as northern states. We exclude the erstwhile state of Jammu & Kashmir due to the unavailability of Census data in 1991. The state of Himachal Pradesh is also omitted as it has a smaller population and area, and we only consider the major northern states. The state of Uttarakhand is also small but has been considered as it was part of the state of Uttar Pradesh until 2000. Similarly, the states Jharkhand and Chhattisgarh are also considered as they belonged to the states of Bihar and Madhya Pradesh, respectively, until the year 2000.

[12]In an additional analysis, we allow the temperature coefficients to vary based on whether the district is in southern states of India, where the custom historically has not been prevalent. The results are reported in Appendix C, highlighting no negative effects for southern districts. These results complement the findings of Corno et al. (2020) .

changes in marriage-related female migration. Therefore, we further corroborate our analysis by providing more evidence on the marriage channel using the NSS-EUS survey for the years 1987-88, 1999-2000, and 2006-07. These waves of EUS had modules on migration where the survey respondents, that is, the members of the household, were asked if their current place of enumeration was different from their last usual place of residence. If so, the primary reason for this migration is recorded. Thus, using this data and the survey weights provided by NSS, for each of these waves of EUS, we construct an outcome variable that is the share of the total married working-age female migrants in the district who have undergone intra-district rural-urban migration due to marriage in last ten years. We compute similar measures for intra-district rural-rural migration. The approach is aligned with the measurement of migration variables using the Census data, but here, the variables are refined further to capture only the marriage-related migration by women.

The computation of climate variables stays the same. We then carry out a panel fixed effects regression in a similar manner as before with district-fixed and year effects. The results are presented in Table 3, and findings from regression (3) indicate a negative association between increases in temperature and female rural-urban marriage migration in the northern regions. These findings lend further support to the initial results, suggesting that the declines in female migration observed previously are driven by decreases in marriage-related migration in northern India.

Table 3: Effects of Rising Temperatures on Marriage Migration, NSS Survey, 1987-2007

	Rural-Urban	Rural-Rural	Rural-Urban	Rural-Rural
	Female	Female	Female	Female
	(1)	(2)	(3)	(4)
Temperature	0.009	0.0005	0.030	−0.001
	(0.016)	(0.044)	(0.017)*	(0.053)
Temperature x North			−0.049	−0.007
			(0.023)**	(0.069)
Precipitation	0.001	−0.023	0.022	−0.013
	(0.015)	(0.033)	(0.015)	(0.034)
Precipitation x North			−0.026	−0.070
			(0.023)	(0.059)
District FE	Y	Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	881	881	881	881

Note: The dependent variables in (1) and (3) are the share of married female intra-district rural-urban migrants of working age who have migrated due to marriage in NSS rounds 1987-88, 1999-00, and 2007-08, respectively. The dependent variables in (2) and (4) are the share of married female intra-district rural-rural migrants of working age who have migrated due to marriage in NSS rounds 1987-88, 1999-00, and 2007-08, respectively. The independent variables in (1) and (2) are the decadal averages of annual temperature (°C) and precipitation (in mm). The independent variables in (3) and (4) are the decadal averages of annual temperature (°C) and decadal averages of annual temperature interacted with "North" dummy denoting 1 if the district lies in a major northern state such as Punjab, Uttarakhand, Uttar Pradesh, Haryana, Bihar, Jharkhand, Rajasthan, Madhya Pradesh, and Chhattisgarh, and 0 otherwise. The regression also controls for the decadal average of precipitation (in mm) and decadal average of precipitation interacted with "North" dummy. The standard errors are clustered by district and are in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Moreover, we posit that these declines in marriage migration by women are due to the diminishing agricultural income and wealth from soaring temperatures in rural areas, which, in turn, is dwindling the resources for dowry in northern areas. We provide suggestive evidence of shrinking dowry amounts in rural northern India in figure 4 below. The figure displays a negative relationship between net dowry paid by brides' households and rising temperatures.

IV.I Robustness Tests

In this section, we discuss the robustness tests we implement to further validate our main results regarding female migration rates in Table 1. These tests include adding state-year trends to the regression equation, re-running the analysis using alternative climate variables, and accounting for spatial autocorrelation. The results of robustness tests are attached in Appendix B.

State-year trends. In the first robustness test, we add state-year trends to our regression equation to capture state-specific time trends. The findings, presented in Table A3, show a slight increase in the magnitude of the estimates. Importantly, the direction and statistical significance of the results stay consistent with the initial findings.

Alternative climate variables. We explore the sensitivity of our main results to different measures of temperature and precipitation. For this, we aggregate grid-level weather data to district-level outcomes by calculating the weighted average of all grid points within 100 kilometers of each district's geographic center, using an inverse-square weighting of the distances, as suggested by Taraz (2018) . These results, which are displayed in Table A4, affirm the robustness of the initial findings. The estimates remain statistically significant and consistent with the main results. The magnitudes of the estimated coefficients have increased slightly, indicating that the effects of rising temperature on female migration rates might be even higher than previously estimated.

Conley standard errors. In the third robustness test, we adjust the standard errors to consider possible spatial auto-correlation (Conley, 1999; Hsiang, 2010). We follow Sekhri and Storeygard (2014) and set the distance cutoffs at 100, 150, and 200 kilometers. The results, detailed in Table A5, demonstrate that the estimates remain robust when applying Conley standard error adjustments for these distance cutoffs.

V Mechanisms and contextual factors

In this section, we first investigate the underlying mechanism driving our results. We explain that declines in agricultural yields due to rising temperatures are eroding the wealth of rural households. This, in turn, harms their ability to finance the dowry payments. In addition, we test whether the effects of rising temperatures on female migration rates vary with a contextual factor, such as poor access to credit.

V.I Agricultural Yields and Dowries

As we discussed before, evidence supported by simulation models and observational data shows the detrimental effects of temperature rises on agricultural yields in India. We replicate these findings for the time period of our study and, specifically, carry out an analysis to dissect the effects for the northern districts. We follow Taraz (2018) and Liu et al. (2023) to measure the yearly aggregate agricultural yield at the district level. As detailed in Section III, we use the VDSA data on agricultural yields and compute the aggregate agricultural yield at the district level for the time period of our analysis, 1981-2010. Thereafter, we conduct a panel fixed-effects estimation with the natural logarithm of the aggregate agricultural yields as the dependent variable[13].

As expected, the results in Table 4 show a marked decline in agricultural yields due to rising temperatures.

Table 4: Effects of Rising Temperatures on Agricultural Yields, 1981-2010

	Log Agr. Yield
	(1)
Temperature	-0.065
	$(0.019)^{***}$
Precipitation	0.044
	$(0.010)^{***}$
District FE	Y
Year FE	Y
Observations	8,693

Note: The dependent variables in (1) is the natural logarithm of the aggregate agricultural yields, between the years 1981 and 2010. The independent variables in (1) are the average annual temperature (°C) and precipitation (in mm). The standard errors are clustered by district and are in parentheses.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

How do these declines in agricultural yields translate into lower resources for dowry? The gradual declines in agricultural yields from rising temperatures over the years are decreasing income for agricultural households. However, rural households in India exhibit consumption patterns characterized by habit formation, meaning they slowly adjust their consumption in response to income volatility (Khanal et al., 2019), often delving into their savings to finance consumption. As the agricultural households gradually and disproportionately adjust their consumption to the decreases in agricultural incomes, far fewer resources are left for savings for dowry[14].

The social and economic costs of not being able to get a daughter married are high (Anukriti et al., 2022), which can compel a few agricultural households to possibly overcome these losses in savings by adapting to rising temperatures and finding ways to save enough for dowry. However, this could be specifically more difficult for relatively poorer households[15], who are unable to adapt and fail to save enough. As the cost of an unmarried daughter is high, these households

[13] We execute the following econometric analysis:

$$lnY_{j,t} = \eta_0 T_{j,t} + \eta_1 P_{j,t} + \phi_j + \gamma_t + \epsilon_{j,t}$$

Here, $Y_{j,t}$ is the aggregate agricultural yield in district j and year t. $T_{j,t}$ is the annual average temperature in °C and $P_{j,t}$ is the annual average precipitation in mm in district j and year t. We control for district fixed effects using ϕ_j and year effects through γ_t.

[14] We cannot completely discount that consumption levels are not changing. Studies have discussed the negative effects of rising temperatures on the consumption of rural households (Liu et al., 2023). More likely, both consumption and savings are drying up.

[15] Approximately 70% of the rural population in India has zero to little landholdings (Asher et al., 2022). Research shows considerable linkages between the size of landholdings and access to adaptation to climate change (Baland & Robinson, 2008; Bryan et al., 2009; Deressa et al., 2009; Dustmann & Okatenko, 2014).

save in advance and enable borrowing during the marriage year to finance dowry and wedding expenses (Anukriti et al., 2022).

For this analysis, we identify districts with low access to banks in the baseline year 1980. We determine whether the per capita number of bank branches was below the national median and create a binary variable to represent this condition: it equals 1 when the per capita number of bank branches in a district in 1980 is below the national median and 0 otherwise. The spatial plot in Figure A.4 in the Appendix shows significant variation in access to bank branches across districts in 1980. To assess how it moderates the effect of rising temperatures on female migration rates, we carry out a variation of the regression equation (2). We allow the coefficients attached to temperature to change based on whether the district had poor access to credit at the baseline.

The findings in Table 5 outline that limited access to banks significantly shapes the impact of temperature increases on female migration, both rural-rural and rural-urban, with a more substantial effect on the latter. Our results confirm that the access to credit constraints faced by poor agricultural households prevent them from saving for dowry in advance, challenge securing loans during the marriage years, and perhaps even constrain their ability to adapt to rising temperatures. This, in turn, is intensifying the effect of climate change on the marriage prospects of rural women and the marriage-related migration by them. In magnitude terms, these results show that a 1 °Celsius increase in temperature is associated with a 25% decline in female rural-urban migration in districts with low access to banks and a 19% decrease in female rural-rural migration. The lower magnitude of the declining effect associated with rural-rural female migration hints that rural households in districts with poor access to credit are perhaps resorting to marrying their daughters to other rural areas rather than to urban areas by offering lower dowry amounts.

Additionally, we adjust the analysis to examine the effects specifically in the northern districts. The findings are depicted as results of regressions (3) and (4) in Table 5. Unsurprisingly, we observe that the diminishing effects are driven by northern districts and, more so, by northern districts with limited access to credit at the baseline.

women's welfare (Calvi & Keskar, 2021, 2023). On a different note, Chiplunkar and Weaver (2023) 's recent study suggests that, as India's literacy rate is rising sharply, the dowry custom might contract in the future as women's literacy improves and the supply of educated grooms increases. However, other studies present a contrasting view, arguing that dowry custom would continue to play a fundamental role in marriages in India in the coming years (Beauchamp et al., 2017). This is because grooms increasingly seek educated women as brides but not highly educated. Thus, highly educated women may face challenges in finding suitable grooms and might need to offer substantial dowries for marriage.

Nonetheless, an important policy recommendation based on the findings is improving access to credit in rural areas. Burgess et al. (2014) have demonstrated how access to banks can reduce mortality in rural areas during summers in India. Banks facilitate consumption smoothing and provide loans at manageable interest rates, which can be crucial for financing dowry resources and adapting to climate change. For instance, farmers can obtain credit to invest in adaptation strategies such as irrigation, crop diversification, mechanization, and purchasing seeds. Additionally, banking access also offers resources for migration to cities. This has been highlighted by Bryan et al. (2014) , who found that offering small loans to subsistence-based farmers and job matching can reduce their aversion to migration. However, it is critical that financial inclusion policies are gender-sensitive. This ensures that women have equal opportunities to migrate for education or work, in addition to traditional marriage-related migration. Mehrotra and Parida (2017) argue that women's labor force participation is likely to increase as more women attain education. Yet, this progression depends on women's mobility and sense of safety when moving out. Therefore, policy interventions are required to first encourage the acceptance of women migrating for reasons other than marriage, such as education or employment opportunities, and make destination places safer. Furthermore, as suggested by Anukriti et al. (2022) , policies that liquidate dowry, such as financial inclusion policies that mitigate barriers to savings and induce savings behavior in poor rural households, can evade distress borrowing at exorbitant rates from informal sources during the time of marriage.

Finally, our study highlights the urgent need for state interventions to transform rural economies to be more resilient to climate change. Infrastructure programs, like road construction and electrification that connect rural areas to cities, can enhance access to agricultural markets, reduce information costs, and bolster the uptake of adaptation means by agricultural households.

VII Conclusion

The literature on climate change and migration in developing countries has expanded significantly in recent years. However, there remains a notable gap in understanding the specific impacts on women, who often constitute a major share of internal migrants due to the patrilocal structure prevalent in these regions. Although female migration appears non-economic, there are often various economic channels underlying the marriage of a woman in these countries. In our study, we demonstrate that rising temperatures are decreasing female migration rates in India, specifically female marriage-related migration in northern regions. We contend that this is due to the historical prevalence of dowry in northern India and identify decreasing agricultural yields from soaring temperatures as an underlying mechanism. In particular, we show that the reductions in agricultural yields are shrinking resources to finance dowry. We further note the declining effects are much more pronounced in regions with limited access to credit.

It should be noted that various norms surrounding marriages, other than dowry, in the country remained largely unchanged during the time period of our analysis, which upholds the validity of our results. For instance, marriages continue to be arranged by parents with little say from the daughters in rural areas, a trend that persisted despite societal modernization and contrasts with predictions of theories of family change (Allendorf & Pandian, 2016; Banerji & Deshpande, 2021). Additionally, marriages continue to occur within the same caste and religion (Allendorf & Pandian, 2016; Banerjee et al., 2013). Comparing 1984 and 2009, Kamble et al. (2014) show that despite massive cultural shifts, mate preferences in India have persisted, including the preference for grooms with higher perceived economic security by women. These factors further support the results of our study.

Nevertheless, our study has important limitations. Firstly, although we have provided ample evidence showing that most marriages occur within short distances of native places during the time period of our study and within districts, we cannot completely discount the possibility that climate change might be inducing rural households to marry off their daughters to far-off places with different weather as a risk diversification mechanism (Rosenzweig & Stark, 1989). This is an issue that could not be explored in our study due to the unavailability of data from the Census of India and NSS, but it is a useful avenue for future research. Secondly, it might be the case that village endogamy, or marrying within one's own village, became more common during our analysis period, which could not be captured through migration rates. In this context, we rely on evidence provided by Mazumdar et al. (2013) , who, based on their primary survey between 2009 and 2011, highlight a contrasting trend of increasing exogamous marriages in previously endogamous regions and note an overall rise in exogamous marriages among younger cohorts.

Our study makes a crucial contribution to the literature on the gendered economic effects of climate change and underlines how climate change can potentially increase economic inequalities by disproportionately affecting women. The findings of our study can be utilized in future studies to investigate further the broader dynamics of climate change impacts on marriages, female migration, and gender outcomes in other developing countries where marriage-related payments such as dowry and bride price are widespread. Moreover, future research should also examine how such traditional gender norms in interaction with climate change are shaping other economic outcomes, such as educational attainment, and social outcomes, such as domestic violence, for women in developing countries.

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Malaria and Human Capital Accumulation

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Abstract

This paper investigates how malaria affects education in the tropics. To do so, we combine unique administrative data on the test scores of over a million primary school students in a high-stakes national exam with georeferenced data on malaria prevalence in Rwanda. The identification strategy exploits spatial and temporal variations in malaria suitability induced by weather conditions. Findings from the paper show that exposure to malaria is associated with lower student performance. Female students are disproportionately affected, thus widening the gender gap in education. Abatement strategies, such as the uptake of insecticide-treated nets (ITNs) help to moderate the effects of malaria on students' performance.

JEL classification: O15, I24, I25

Keywords: malaria, test scores, human capital, gender inequality, Rwanda

1. Introduction

The COVID-19 pandemic highlights the negative socioeconomic impacts associated with diseases. Meanwhile, malaria has been impacting the health and economic outcomes of people in the tropics for several millennia (Gallup and Sachs 2001; Akyeampong 2006; Depetris-Chauvin and Weil 2018). The disease is endemic in 94 countries, with Sub-Saharan Africa accounting for the highest concentration of countries at risk of malaria.[1] In 2022 alone, 249 million cases of malaria infection and 608,000 malaria-related deaths were recorded. About 95 percent of these deaths were recorded in Africa (World Health Organization 2023).

Aside from the worrying morbidity and mortality rates associated with malaria, available evidence suggests economic losses due to malaria are immense, amounting to more than US$12bn every year for the African continent over recent decades (Cervellati et al. 2022). Further, the extant literature suggests

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1 Almost every country in Sub-Saharan Africa is at risk of malaria.

documented evidence of a negative impact of malaria on labor productivity and hence wages (Bleakley 2010; Hong 2011). This is likely to reduce the expected returns to schooling and therefore reduce households' incentive to educate their children. On the other hand, malaria can also reduce the opportunity cost of schooling through low childhood wages (Bleakley 2010). These suggest that the nexus between malaria and schooling decisions is ambiguous depending on the relative effects of expected future earnings vis-a-vis childhood wages.

In this paper, we focus on the short-run effects of exposure to malaria on the performance of primary-school students in Rwanda. We also examine the role of coping strategies, such as the use of insecticide-treated bed nets, in mediating the impact of malaria on students' performance. To estimate the effect of malaria on students' performance, we utilize an administrative dataset on test scores of the universe of primary-school students in Rwanda participating in a high-stake national exam between 2012 and 2017. We link these data with granular spatial data on malaria suitability at a spatial resolution of $0.1^\circ \times 0.1^\circ$ ($\approx 11 \times 11$ km at the equator). Our identification strategy relies on naturally occurring variations in malaria suitability across space and time, while accounting for a battery of fixed effects and controls.

Three main findings about the effects of malaria on students' test scores emerge from the study. First, malaria reduces students' test scores. A one-standard-deviation increase in malaria suitability decreases students' aggregate, STEM, and non-STEM test scores by 0.31, 0.3, and 0.29 standard deviations, respectively. The effects are precisely estimated as they survive a battery of fixed effects.

Secondly, the findings show that malaria contributes to widening the gender gaps in educational attainment. A one-standard-deviation increase in malaria suitability reduces girls' aggregate, STEM, and non-STEM test scores by 0.037, 0,045, and 0.03 standard deviations, respectively, relative to boys. Thus, even though mosquitoes do not discriminate in biting men and women, their effects on students' performance are unequally distributed across gender. The role of sociocultural norms in labor allocation at the household level is a plausible explanation. For instance, in many societies women and girls bear a greater burden of caregiving at the household level, hence if a household member is affected by malaria, girls may be required to temporarily pause schooling to take care of the family member until s/he recovers while boys are allowed to attend school. These interruptions can hinder learning and consequently widen performance gaps between male and female students.

Finally, we find absenteeism or low school attendance to be a likely channel through which malaria affects student test scores. We combine our data on malaria prevalence and survey measures of school attendance to examine the association between malaria and the probability of school-aged children attending school in Rwanda. The results show that malaria incidence is positively associated with absenteeism. That malaria reduces school attendance and consequently student performance underscores the direct and indirect effects of the health impact of malaria on children. They arise from children falling ill themselves or having to stay home to care for other sick family members at the expense of schooling.

We perform several robustness checks. First, our short-term estimates could be biased by long-term adaptation to malaria and selective migration. In a long-term context, adaptation and migration could be mechanisms that mediate the relationship between malaria and human capital accumulation. However, in a short-run context, adoption and migration could be sources of potential bias. For example, places with a long history of malaria prevalence can develop coping strategies to mitigate the effects of malaria compared to people in places with intermittent exposure to malaria parasites, while some places may never be exposed to malaria, and therefore people in such areas would never have to worry about contracting the disease. This creates the classic problem of always takers and always defiers. Furthermore, malaria could induce periodic migration. If families of "high-ability" individuals migrate out of areas with high malaria-suitability zones to less suitable zones, then the estimated negative effects may not necessarily reflect the true effect of malaria, but rather high-performing students are disproportionately located in low-malaria zones, which could lead to overestimation of the impact. To assess the potential biases of adaptation and migration, we conduct a two-tail trimming exercise where we progressively exclude observations that always fall in high- or low-malaria zones along the malaria-suitability distribution and estimate using the remaining observations. Places with historically high malaria incidences are likely sources of malaria-related migration while places with historically low incidence of malaria are likely destinations of malaria-related migration. The estimated effect of malaria on all test scores persistently remains negative even when we remove 95 percent of the distribution of malaria suitability and the magnitude of the estimated effect is always comparable to the full sample estimation.[2] In addition, if malaria suitability predicts other diseases, it might bias the estimated effects. To examine this, we assess the relationship between malaria suitability, and diarrhea and cough which are common symptoms of communicable diseases prevalent in Rwanda. We find no evidence that malaria suitability predicts diarrhea or cough (supplementary online appendix S1).

This paper makes two main contributions to the literature. First, the finding that malaria reduces test scores speaks directly to the strand of the literature showing the socioeconomic impacts of disease environments. There is a growing interest in understanding the role of disease environment in explaining differences in comparative development (Gallup and Sachs 2001; Acemoglu, Johnson, and Robinson 2003; Bleakley 2007; Alsan 2015; Depetris-Chauvin and Weil 2018), productivity (Baldwin and Weisbrod 1974; Bleakley 2010), and human capital (Cutler et al. 2010; Bleakley 2010; Archibong and Annan 2017). While malaria is endemic in many countries, a good number of countries have in the last century successfully eradicated malaria. These countries provide a unique quasi-natural experiment for evaluating the long run impact of malaria by comparing populations exposed to malaria with comparable cohorts that were not exposed to malaria (Bleakley 2010; Cutler et al. 2010; Lucas 2010; Shih and Lin 2018). Lucas (2010) for instance, examines the long-run impact of malaria eradication in Sri Lanka and Portugal on educational attainment and finds consistent evidence that the eradication had a positive effect on educational attainment and literacy in both countries. Despite the high prevalence of malaria in Africa, there are few studies, to the best of our knowledge, that provide a robust analysis of the socioeconomic consequences of malaria, particularly in terms of education (Burlando 2012; Kuecken, Thuilliez, and Valfort 2014). A plausible reason for this under-representation of Africa in studies on the economic consequences of malaria for instance could be the challenges associated with causal identification of impacts and data challenges. We attempt to fill this gap by leveraging a unique administrative dataset on the universe of primary-school students in an African country linked with georeferenced data on malaria suitability.

Next, this paper adds to the fledgling strand of literature on the disease origins of gender inequality (Archibong and Annan 2017, 2020). Even though the socioeconomic consequences of diseases are well established, little is known about the effects on inequality, particularly along the lines of gender. Evidence from Archibong and Annan (2017, 2020) shows that diseases can contribute to widening the gender gaps in education in developing countries, with factors such as the indirect economic cost of disease burden which induce households to marry off their daughters at an early stage, and endogenous social norms as likely operative channels. Our findings of a malaria-induced gender gap in test scores speak to the latter channel. Social norms that require women (and girls) to spend time at home taking care of sick family members may lead to interruptions in school attendance, learning, and consequently student performance.

The rest of the paper is structured as follows. Section 2 outlines the data used in the analysis while Section 3 describes our identification strategy. We present and discuss our results in Section 4. We share the robustness checks and mechanisms in Sections 5 and 6, respectively. Section 7 concludes the paper with a summary of findings.

2 We check sensitivity of the main result by excluding the observations lying within the remaining 5 percent of the distribution of malaria suitability and we find that excluding these observations does not significantly change our main result.

2. Data

The main dataset used in the analysis is an administrative dataset on students' test scores in a primary-school national exam matched with georeferenced data on malaria suitability in Rwanda. We complement the analysis with several datasets as discussed in the following subsections.

2.1. Test Scores

Our test scores data comes from the Rwandan Educational Board (REB), which administers the primary-school national examinations in the country.[3] Primary-school students in Rwanda undertake six years of education with a mandatory national (exit) exam in five subjects (mathematics, science, social studies, Kinyarwanda, and English) after completing the sixth grade. Performance in this national exam is used as the metric for progression into middle schools (lower secondary schools). Although all sixth graders are eventually promoted into middle schools, best-performing students in the national exams are automatically given places in boarding schools while the low-performing students are promoted into the same schools in what is commonly known as 12 years basic education (12YBE). Thus, our test score data are repeated cross-sectional data at school level, and contain information on students who sat for the exam during the sample period.

This dataset contains the universe of primary-school students, approximately 1.37 million, who sat for the national exams between 2012 and 2017. The dataset contains two measures of student outcomes: (a) the standardized test scores of each student for the respective subjects;[4] and (b) the corresponding grade points, ranging from 1 to 9, with 1 being the best grade and 9 the lowest grade. These grade points are usually aggregated to determine a student's overall performance in the exam, such that the top-rank student obtains an aggregate of 5 while the lowest-rank student obtains an aggregate of 45. Since we are interested in students' aggregate performance in all subjects, we use the grades in the respective subjects to generate a standardized aggregate grade point. Specifically, we create a modified aggregate grade point by reversing the scale (ranking) of the grades such that 1 means poor and 9 means excellent. We then aggregate these (rescaled) grades to obtain our "modified" aggregate grade points ranging from 5 to 45, with grades 5 and 45 representing the lowest and highest grades, respectively. We also compute the total grade points for STEM (mathematics and science) and non-STEM (social studies, Kinyarwanda, and English) subjects. For ease of interpretation and to ensure consistency with the test scores of the individual subjects, we standardized the grade points for the aggregate, STEM, and non-STEM subjects. Figure 2 shows the spatial variations in average test scores across locations.

Finally, the dataset also contains information on the gender and age of students. We complement the data with geo-coordinates of all primary schools in the country as shown in fig. 3.

2.2. Malaria Suitability

There are two main malaria parasites endemic in Africa, namely, *Plasmodium falciparum* and *Plasmodium vivax*. The prevalence of these parasites however varies across locations. In Rwanda, for instance, the endemic malaria parasite is *Plasmodium falciparum*. The prevalence of malaria depends largely on the malaria ecology, i.e., the conditions suitable for the survival of the malaria parasites which in turn are largely influenced by weather such as temperature. Therefore, weather conditions influence the survival and spread of malaria across locations and time.

3 Since 2021, the responsibility for administering primary-school national exams has been given to the National Examination and School Inspection Authority (NESA).

4 Each year's test scores for the five subjects are aggregated into aggregate, STEM (math and science) and non-STEM (social studies, Kinyarwanda, and English) and are standardized with mean 0 and standard deviation 1. Hence, each data point measures the standard deviations from the mean.

40° C, considered unsuitable between 16° C and 18° C, and highly favorable above 22° C (Craig, Snow, and le Sueur 1999; Sachs and Malaney 2002).

Seasonal temperature variation is a main determinant of malaria transmission rates. At lower temperatures, most mosquito vector species become inactive or significantly reduce biting activity (Sachs and Malaney 2002). Conversely, higher temperatures accelerate mosquito development and shorten the interval between blood meals, increasing contact between mosquitoes and hosts. However, at extremely high temperatures, mosquito survival rates decrease significantly (Gething et al. 2011).

We construct a relative measure of temperature suitability to malaria transmission following the approach detailed in Gething et al. (2011).[5] Then a relative measure of malaria suitability $Z(T)$ is specified as[6]

$$Z(T) = \frac{e^{-g(T)n(T)}}{g(T)^2}, \tag{1}$$

where $g(T)$ is vector survival in terms of daily death rate,

$$g(T) = \frac{1}{(-4.4 + 1.31T - 0.03T^2)},$$

and $n(T)$ is the sporogony duration required for the malaria vector to be infectious. Sporogony duration is a function of a fixed number of days φ, the minimum temperature required for development $T_{\min}$, and average monthly temperature T:

$$n(T) = \frac{\phi}{(T - T_{\min})}.$$

We apply the widely used parameter values in the malaria literature $\varphi = 111$ and $T_{\min} = 16°$C (Gething et al. 2011).[7]

We construct the malaria-suitability index at a spatial resolution of $0.1° \times 0.1° \approx 11\ \text{km}^2$ grid-cell level using temperature data from the Copernicus Climate Data Store (Muñoz Sabater 2019). For grid cells that experience a monthly temperature of below 16° C and above 40° C, we replace the suitability index with zero. Then we compute yearly averages that match with our test score data. For ease of interpretation and without loss of generality, we standardize the malaria-suitability index to zero mean and unit standard deviation. The spatial granularity of our malaria-suitability index enables us to explore the changes in malaria suitability across space and time at a very high spatial resolution. This is important, particularly in the case of small countries such as Rwanda.

Admittedly, another key factor in determining malaria suitability is precipitation. However, once the minimum precipitation requirement is met, precipitation does not influence malaria suitability to the same extent as temperature does. In addition, a dry month can be suitable for transmission when previous months have received sufficient precipitation. Geo-climatic conditions in Rwanda enable the country to receive sufficient rainfall throughout the year to be suitable for mosquito reproduction and malaria transmission. We test Rwanda's precipitation suitability for malaria transmission using the conditions suggested by Tanser, Sharp, and Le Sueur (2003). The conditions include (a) three-month moving average precipitation of at least 60 mm, and (b) at least one month in three months (the current month and the previous two months) has precipitation of at least 80 mm. We find that the average monthly precipitation in a grid cell is 1402.4 mm (sd 1315 mm). Similarly, for the duration of our sample period, on average 98.7 percent of months in grid cells satisfy condition (a), 99.2 percent satisfy condition (b),

5 The malaria suitability is a relative measure, meaning that a place with a suitability index of 10 is twice as suitable for malaria transmission as a place with a suitability index of 5.

6 For derivation of the functional form, see Gething et al. (2011).

7 The resulting theoretical relationship between malaria suitability and temperature is shown in supplementary online appendix fig. S3.1.

and 98.7 percent satisfy both conditions. This suggests near-universal malaria suitability across locations in the country, which is contrary to administrative data from the country indicating places with little or no malaria endemicity. Therefore, we argue that temperature suitability is plausibly the most accurate measure of determining the conditions for malaria in the country, and hence its use in this paper.[8]

Finally, using the geo-location of the schools in our exam dataset, we spatially match our malaria-suitability index to the schools to measure malaria exposure at a given location and time. An implicit assumption in this approach is that students live in the same grid cell as where a school is located. The wide geographic reach of primary schools and the fact that most primary-school students live within 2–7 km from their school provides reasonable support for this assumption.[9]

2.3. Complementary Datasets

We complement our analysis with additional datasets. First, we leverage household data from Rwanda's Integrated Household Living Conditions Surveys (EICVs) to examine potential mechanisms that explain the effects of malaria on students' performance. The EICV is a nationally representative survey of households in Rwanda and covers a wide array of issues related to the socioeconomic welfare of households in the country. Three rounds of the survey conducted in 2010/2011, 2013/2014, and 2016/17 were used. Specifically, we use data on school attendance (absenteeism) of school-aged children (6–18 years old), among others.

In addition, we used data on the location of health centers in Rwanda from the Rwanda Biomedical Center (RBC), as well as the uptake of insecticide-treated (bed) nets (ITNs) from the Malaria Atlas Project (Bhatt et al. 2015). Data from the Demographic and Health Surveys (DHS) are also used in the analysis. Finally, in our regression analysis, we control for baseline characteristics such as agricultural suitability (Fischer et al. 2021), population density (Center for International Earth Science Information Network - CIESIN - Columbia University 2018), nightlights (Li et al. 2022), and road density Meijer et al. (2018), interacted with time trends. Figure 4 shows the spatial variations in the malaria-suitability index and uptake of ITNs in the country.

3. Empirical Strategy

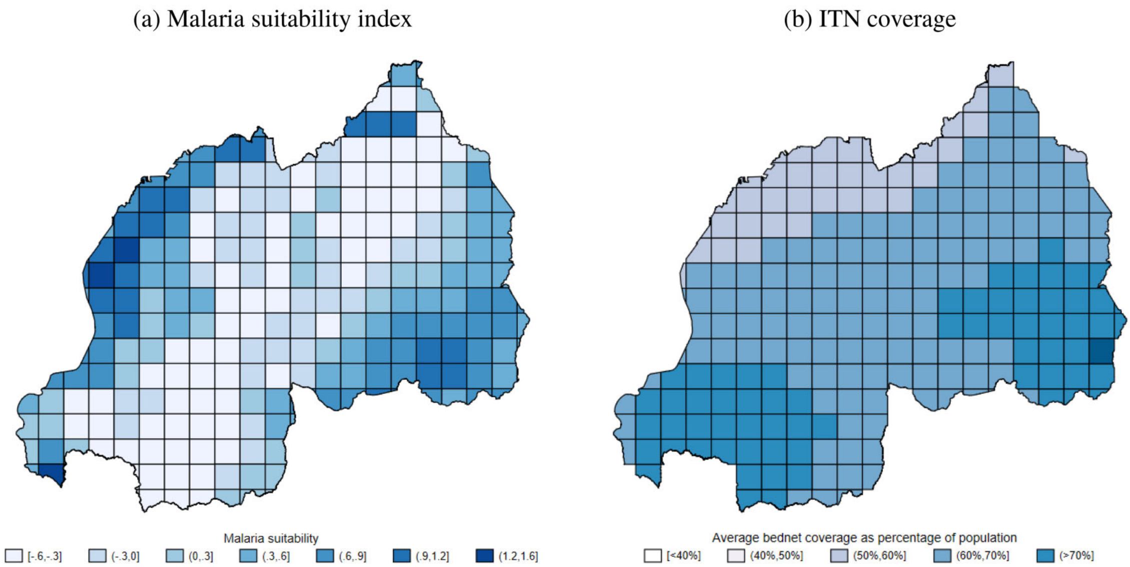
A major challenge to causal identification of the impact of malaria infection on socioeconomic outcomes such as students' test scores is the plausibly endogenous relationship between these outcomes and malaria infection. Factors such as the quality and access to sanitation, access to health care, and coping strategies (e.g., utilization of treated bed nets) are likely to confound the relationship. Another concern relates to measurement errors in data on malaria incidence, given the lack of granular administrative data on malaria incidence in the study area.

To address these challenges, our research design relies on a reduced-form model that leverages naturally occurring variations in the conditions suitable for malaria transmission to estimate the effect of malaria on students' performance. Geographic and weather conditions are the key determinants of malaria ecology. Highly suitable malaria zones tend to record a high incidence of malaria, as mosquitoes and malaria parasites thrive and reproduce effectively in these areas. Given the importance of weather conditions on the malaria ecology, malaria suitability changes over time are driven largely by the changing climate. Over time, even places that had low malaria suitability can experience increased malaria infections and vice versa. Thus, we leverage the changes in malaria suitability to identify the impact of malaria on our

8 As shown in supplementary online appendix fig. S3.2, using malaria incidence data from the Malaria Atlas Project (Weiss et al. 2019), our measure of malaria suitability is highly correlated with incidence. In addition, supplementary online appendix fig. S3.3 also shows relatively strong temporal variations in the suitability index.

9 See for example Kwakwa (2023).

Figure 4. Malaria Suitability and ITN Coverage



Source: Figure based on malaria suitability constructed using data from Muñoz Sabater (2019) and ITN coverage from the Malaria Atlas Project (Bhatt et al. 2015).
Note: Panel (a) plots constructed malaria transmission suitability in each grid cell averaged across years and standardized with mean 0 and standard deviation 1. Panel (b) plots the average insecticide tread bed net coverage per grid cell.

outcomes. Consequently, our reduced-form equation is specified as follows:

$$Y_{isgt} = \phi \times \text{Suitability}_{gt} + \beta X_{isgt} + \alpha_s + \lambda_t + \text{YOB}_{\text{FE}} + \epsilon_{isgt}, \tag{2}$$

where Y_{isgt} is the test scores of student i in school s located in grid cell g who sat for the exam in year t. The variable Suitability_{gt} is the standardized malaria-suitability index in grid cell g at time t, and X_{isgt} is a vector of controls such as gender of student i, average annual temperature, and total annual precipitation at the grid-cell level. To control for underlying differences in socioeconomic development across locations that may be correlated with our malaria-suitability measure, we include as covariates, baseline characteristics such as crop suitability,[10] population density, nightlights, and road density, interacted with time trends. The variables α_s, λ_t, and YOB_{FE} also represent school, exam-year, and birth-year fixed effects, respectively. The school fixed effects (the lowest spatial unit in our data) absorb time-invariant (observable and unobservable) differences across schools (and locations) that could influence students' performance. These include school quality, and location-specific characteristics such as altitude and proximity to water bodies. Exam-year fixed effects also absorb contemporaneous correlates of student performance. This includes, for instance, changes in local economic conditions (income) that could be trending with student performance, thereby conflating our estimates. Finally, birth-year fixed effects absorb age-specific correlates of student performance such as grade progression, and in utero or postnatal shocks with potentially long-term impact on cognitive performance. For robustness, we present additional estimates using a battery of fixed effects including province × year, school × gender, and school type × birth-year fixed effects.[11]

The validity of our research design relies on the assumption that variations in malaria suitability are as good as random. The fact that variations in our malaria-suitability measure are primarily driven by temperature shocks, which are arguably exogenous, provides some comfort. However, given that our

10 Coffee, tea and maize.

11 School type refers to whether the school is owned and managed by the government, by religious institutions, or is in private ownership.

malaria measure as described in Section 2.2 is a nonlinear function of temperature, we face the challenge of disentangling the pure effects of malaria on test scores from the "direct" effect of temperature on test scores (Park et al. 2020; Srivastava, Hirfrfot, and Behrer 2024). To address this challenge, we directly control for temperature in our regression, thereby isolating the linear relationship between temperature and student performance from conflating our results. Still, some concern remains that our estimation specifications do not allow for controlling the nonlinear effects of temperature, a limitation we discuss in Section 7.

Another source of concern is the so-called selective migration in response to malaria ecology. Given the well-established fact of a relatively high health burden associated with malaria, it is reasonable to assume that without mobility and resource constraints, households would migrate from high- to low-malaria-endemic zones. However, resource and mobility constraints may favor the migration of certain groups of people (e.g., resource-endowed and high-ability households). This could potentially create selection bias in the sample such that low-income/ability students may remain in malaria-endemic zones and vice versa. As a result, our estimates would likely be confounded by the effects of malaria-induced migration. For instance, a negative effect could arise from comparing high-ability students in low-malaria zones with low-ability students in high-malaria zones. In Section 5, we address this issue and show that our estimates are unlikely to be influenced by malaria-induced migration.

Thus, conditional on the set of fixed effects, our coefficient of interest, ϕ, measures the intent-to-treat estimate of changes in malaria suitability on students' performance. In the baseline specification, standard errors are clustered at the grid-cell level to account for within-cluster correlation. The results remain robust to alternative clustering including spatially correlated standard errors and clustering at subnational levels.

4. Results

4.1. Main Results

We begin by presenting our reduced-form estimates of the effect of changes in malaria-suitability conditions on aggregate test scores in all subjects as well as in STEM and non-STEM subjects. As shown in table 1, we estimate variant specification alternating between the various sets of fixed effects, with column 2 being our preferred specification, where we control for school location, exam-year, and birth-year fixed effects in addition to the gender of the students. Across the various specifications, the results remain quantitatively and qualitatively similar.

We find that a 1-standard-deviation increase in malaria suitability is associated with a 0.32-standard-deviation decrease in aggregate test scores (column 2). The effects on performance in STEM and non-STEM subjects are relatively similar: a decline of about 0.3 standard deviations, respectively. Beyond the effects on aggregate scores, we also present estimates of the effects of malaria on students' performance in the respective subjects in supplementary online appendix table S2.1, and find a similar negative association between malaria suitability and test scores. Overall, the results unanimously show a negative effect of exposure to malaria conditions on students' performance in high-stake exams.

4.2. Does Malaria During Exam Season Matter?

Malaria morbidity could affect the cumulative learning process as well as students' exam performance contemporaneously. Exposure to malaria in the exam month or the few months before the exam could hinder students' preparation and their ability to effectively participate in the exam. We assess whether the effect of malaria on test scores measured in our main regression result pertains to cumulative effect or exposure in the exam month using a reduced-form regression of standardized test scores on malaria

Table 1. Malaria and Student Performance

	Dep. var.: Standardized test scores				
	(1)	(2)	(3)	(4)	(5)
			Panel A: Aggregate		
Malaria suitability	−0.215**	−0.316***	−0.274***	−0.318***	−0.313***
	(0.083)	(0.078)	(0.090)	(0.077)	(0.077)
			Panel B: STEM		
Malaria suitability	−0.204**	−0.300***	−0.264***	−0.303***	−0.296***
	(0.081)	(0.079)	(0.094)	(0.079)	(0.079)
			Panel C: Non-STEM		
Malaria suitability	−0.199**	−0.299***	−0.262***	−0.302***	−0.297***
	(0.085)	(0.076)	(0.088)	(0.076)	(0.075)
Controls	No	Yes	Yes	Yes	Yes
Fixed effects					
School	Yes	Yes	Yes	No	Yes
Year	Yes	Yes	No	Yes	Yes
YOB	Yes	Yes	Yes	Yes	No
School × YOB	No	No	No	No	No
Province × Year	No	No	Yes	No	No
School × Female	No	No	No	Yes	No
School type × YOB	No	No	No	No	Yes
Obs	1,038,792	1,038,792	1,037,147	1,038,788	1,038,792

Source: Regression results are based on test score data from the Rwandan Educational Board, malaria suitability constructed using data from Muñoz Sabater (2019), and control variables extracted from various datasets described in Section 2.3.

Note: The table reports the effect of malaria on test scores following equation (2). Each cell within each panel presents results from separate regressions. Dependent variables indicated as panel titles are aggregate, STEM and non-STEM test scores standardized with mean 0 and standard deviation 1. Standard errors in parentheses are clustered at grid-cell level. YOB is year of birth. *, **, and *** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

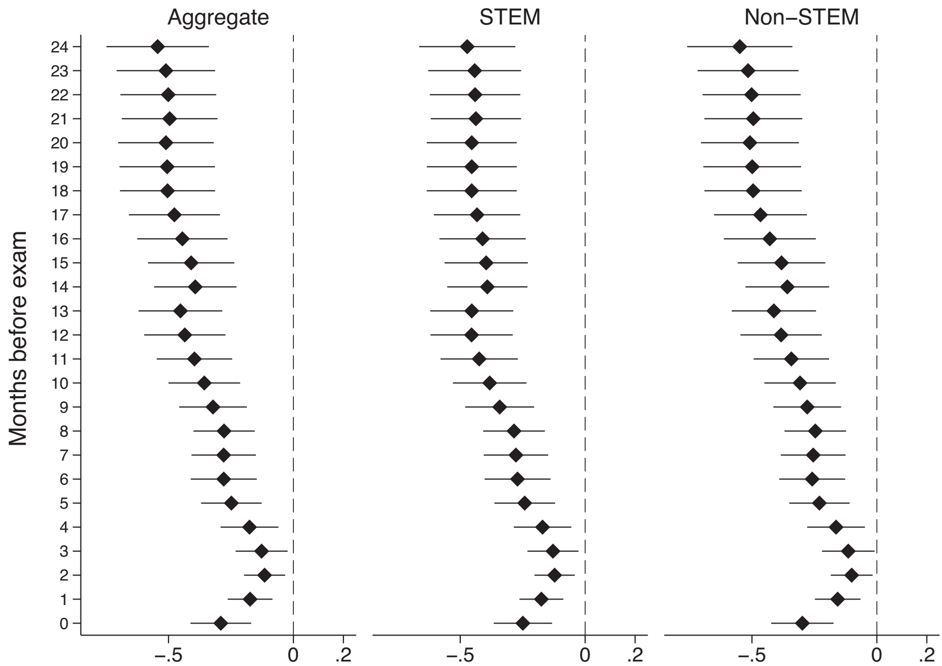
transmission suitability in the exam month as expressed in equation (3):

$$Y_{isgt} = \phi \times \text{MonthSuitability}_{gt} + \beta X_{isgt} + \alpha_s + \lambda_t + \text{YOB}_{\text{FE}} + \epsilon_{isgt}, \quad (3)$$

where $\text{MonthSuitability}_{gt}$ is the standardized malaria-suitability index in grid cell g at the exam month of time t. Other variables remain as defined before. The exam is often conducted in the last week of October and/or the first week of November as shown in supplementary online appendix table S2.2. The latest date an exam is conducted in any given year in our sample is November 7. We take October as month zero and estimate equation (3) using the malaria suitability in the exam month and up to 24 months before the exam in separate regressions. Malaria suitability in the months before the exam is the moving average of the months before the exam considered including the exam month.

We plot the result in fig. 5. The results show that increased malaria suitability in the exam month is associated with lower student performance in the exam. The effect of the average suitability in the 1st to 4th months before the exam is below the effect of the exam month. Interestingly, we find that the effect of the average malaria suitability in the 5th to 24th months before the exam is not only negative and statistically significant but also increasing, in absolute terms. The observed pattern in the effects of changes in malaria suitability over the period leading to the exam, as well as the effect of malaria suitability in the exam month, suggests that malaria has both cumulative and contemporaneous negative effects on students' performance. In other words, where changes in malaria prevalence in local communities affect learning, the effect of malaria during the exam period is even higher. This has implications for policies relating to the timing of examinations in malaria-endemic areas and timing of prevention and treatment administration targeting the school-aged portion of the population.

Figure 5. Reduced-Form Estimates Using Average Malaria Suitability over Various Months Before Exam



Source: Regression results are obtained using data on test scores from the Rwandan Educational Board, malaria suitability constructed using data from Muñoz Sabater (2019), and control variables extracted from various datasets described in Section 2.3.

Note: The figure shows the effect of malaria on test scores during exam month and the months before the exam. Each point comes from a separate regression where malaria suitability is the average of the months (including the exam month) indicated on the vertical axis starting from the exam month. Dependent variables are test scores indicated on top and standardized with mean 0 and standard deviation 1. The regressions include control variables, and year, school, and year-of-birth fixed effects. Horizontal lines indicate 90 percent confidence intervals.

Further, given the significant effect of malaria suitability in the exam month, we also examine whether this result is robust to different combinations of fixed effects. The results presented in table 2 show that the estimated effects remain largely consistent when accounting for alternative fixed effects structures. In column 2, a 1-standard-deviation change in malaria suitability is associated with −0.291, −0.249, and −0.298 changes in total, aggregate STEM, and aggregate non-STEM standardized test scores, respectively.

4.3. Malaria and Gender Gap

In this section, we examine the effects of the disease burden associated with malaria in widening the gender gap in student performance. Although mosquitoes do not discriminate in biting men and women, evidence from the literature points to gender inequalities in the people at risk of malaria and its associated effects (World Health Organization 2023). Even though biologically, pregnant women and children are more vulnerable to malaria, other socioeconomic factors increase the risks of malaria to women (UNDP 2015). Inequalities in resource allocation, intra-household decision-making, and labor allocation are among the primary drivers of gender gaps in the effects of malaria. Many female-headed households in Sub-Saharan Africa (SSA) are low income. This limits the ability of such households to invest in malaria prevention strategies such as treated nets, adequate nutritional intake, quality housing, and even their ability to pay for malaria medications. Also, sociocultural norms in many developing countries influence decision-making in households: women are often not involved in decision-making on a variety of issues including health

Table 3. Malaria and Student Performance: Gender Gap

	Dep. var.: Standardized test scores			
	(1)	(2)	(3)	(4)
	Panel A: Aggregate			
Female	−0.250***	−0.250***	−0.250***	−0.251***
	(0.007)	(0.007)	(0.007)	(0.007)
Malaria suitability	−0.190**	−0.295***	−0.252***	−0.292***
	(0.082)	(0.078)	(0.091)	(0.078)
× Female	−0.038**	−0.038**	−0.038**	−0.037**
	(0.015)	(0.015)	(0.015)	(0.015)
	Panel B: STEM			
Female	−0.266***	−0.266***	−0.266***	−0.267***
	(0.007)	(0.007)	(0.007)	(0.007)
Malaria suitability	−0.175**	−0.275***	−0.238**	−0.272***
	(0.081)	(0.080)	(0.094)	(0.080)
× Female	−0.046***	−0.045***	−0.046***	−0.045***
	(0.015)	(0.015)	(0.015)	(0.015)
	Panel C: Non-STEM			
Female	−0.226***	−0.226***	−0.226***	−0.226***
	(0.007)	(0.007)	(0.007)	(0.007)
Malaria suitability	−0.178**	−0.282***	−0.244***	−0.280***
	(0.084)	(0.076)	(0.088)	(0.076)
× Female	−0.031**	−0.031**	−0.031**	−0.030**
	(0.015)	(0.015)	(0.015)	(0.015)
Controls	No	Yes	Yes	Yes
Fixed effects				
School	Yes	Yes	Yes	Yes
Year	Yes	Yes	No	Yes
YOB	Yes	Yes	Yes	No
Province × Year	No	No	Yes	No
School type × YOB	No	No	No	Yes
Obs	1,038,792	1,038,792	1,037,147	1,038,792

Source: Regression results are based on test score data from the Rwandan Educational Board, malaria suitability constructed using data from Muñoz Sabater (2019), and control variables extracted from various datasets described in Section 2.3.

Note: The table reports the disproportionate effect of malaria on female students' test scores. Each column within each panel presents results from separate regressions. Dependent variables indicated as panel titles are aggregate, STEM, and non-STEM test scores standardized with mean 0 and standard deviation 1. "× Female" indicates interaction between Malaria suitability and Female which is an indicator variable equal to 1 if the student is female and 0 otherwise. Standard errors in parentheses are clustered at grid-cell level. YOB is year of birth. *, **, and *** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

performance in diverse ways. As shown in table 4 (and more details across individual subject results in supplementary online appendix table S2.4), we do not find consistent evidence of a rural–urban gap in student performance associated with malaria. These results provide suggestive evidence that malaria affects students in urban schools just as much as students in rural schools.

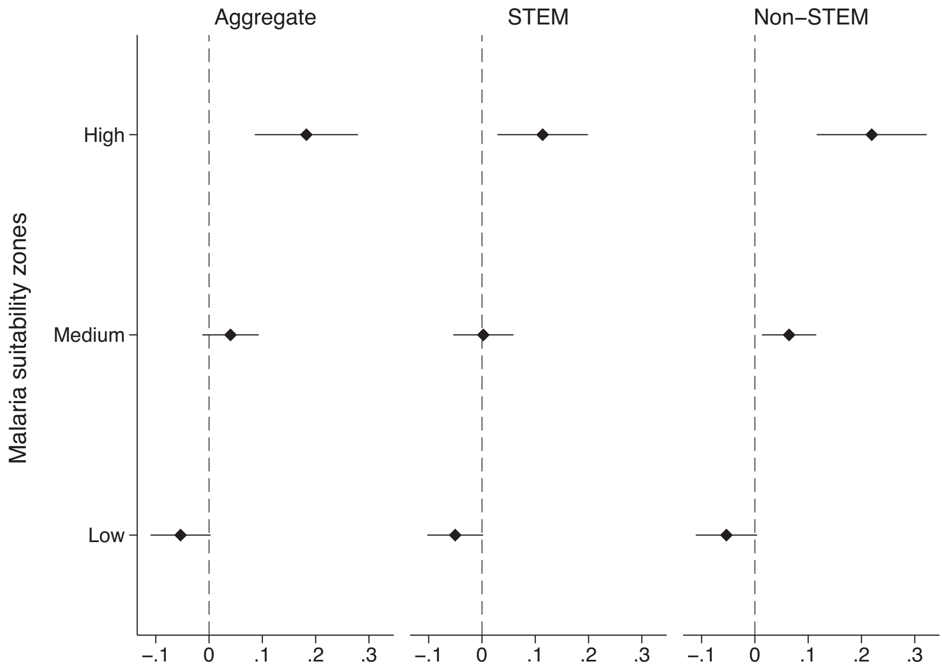
4.5. Role of Access to Insecticide-Treated Nets and Health Care

In this section, we present additional analysis showing how access to treated nets and health care could mediate the effects of malaria on students' performance.

4.5.1. Access to ITNs

The main malaria control strategy recommended by the WHO is the use of ITNs by households in malaria-endemic areas. As a result, significant public and donor funding has been spent on distribution of ITNs to households to limit the incidence and effects of malaria, particularly among pregnant women and

Figure 6. Effect of Treated Bed Nets



Source: Regression results are based on test score data from the Rwandan Educational Board, ITN coverage from the Malaria Atlas Project (Bhatt et al. 2015), malaria suitability constructed using data from Muñoz Sabater (2019), and control variables extracted from various datasets described in Section 2.3.

Note: The figure shows the effect of bed net coverage in different malaria zones on test scores. Each point represents a coefficient from a separate regression on the sample that falls in a malaria-suitability zone. Dependent variables are aggregate, STEM, and non-STEM test scores indicated on top and standardized with mean 0 and standard deviation 1. The regressions include control variables, and year, school, and year-of-birth fixed effects. Horizontal lines indicate 90 percent confidence intervals.

places with low ITN uptake. These effects are statistically and economically significant for aggregate grades (and each subject except mathematics, as shown in the supplementary online appendix fig. S3.4). In medium-malaria-suitability zones, we once again observe a positive effect, albeit the estimates are barely significant statistically, even at a 10 percent error level. In low-malaria zones, we do not observe a significant difference in test scores between students in areas with high ITN uptake and those in areas with low uptake. Overall, the analysis suggests that ITN use is associated with a lower effect of malaria, particularly in high-malaria zones. However, the health literature has well documented the effectiveness of ITNs in mitigating malaria transmission, and their role should not be undermined regardless of our results.

4.5.2. Access to Health Centers

Next, we evaluate the importance of access to health care in the relationship between malaria and test scores. Using geodata on the location of health facilities and schools in Rwanda, we estimate the straight line distance between each school and the nearest health facility. We leverage the distance data and estimate the following doughnut equation:

$$Y_{isgt} = \sum_{d} \phi_d d \times \text{Suitability}_{gt} + \alpha \text{Suitability}_{gt} + \beta X_{isgt} + \alpha_s + \lambda_t + \text{YOB}_{\text{FE}} + \epsilon_{isgt},$$

where $d \in$ 3–6, 6–9, 9–15 km represents the distance bins between each location and the nearest health facilities. The reference group is $d \in$ 0–3 km. Therefore, ϕ_d traces the effects of changes in malaria

5. Robustness Checks

In this section, we present several robustness checks to assess the validity of our estimates.

5.1. Alternative Clustering

Given the spatial dimensions of the malaria-suitability index and the fact that infection spreads across locations, it is plausible that the standard errors are also correlated across space. Therefore, we present additional results by re-estimating the baseline specifications while (a) using the spatially correlated Conley standard errors (Conley 1999) and (b) clustering the standard errors at the municipality (sector) level. The results in supplementary online appendix table S4.1 are largely consistent with our baseline results: finding a statistically significant negative association between malaria and student test scores.

5.2. Adaptation to Local Malaria Environment

The identification strategy of the paper relies on spatial and temporal variations in malaria suitability. In places where environmental conditions are always conducive to malaria, the possibility of adaptation to the disease cannot be ignored. For instance, places with a long history of endemic malaria may develop (local) coping strategies to mitigate the effects of malaria relative to people in places with intermittent exposure to malaria parasites. On the contrary, some places may never be exposed to malaria and hence the people in such areas would never have to worry about contracting the disease unless, of course, they travel to a malaria-endemic zone. This creates the classic problem of "always takers" and "always defiers" for our identification strategy, where always takers are places that are always a malaria zone and always defiers are places that are always not suitable for malaria. To ensure that our results are not driven by these groups, we exclude these groups and re-estimate our analysis.

To do this, the following strategy was used. First, for each of the sample years, we divide the distribution of the malaria-suitability index into 2.5 percentile bins and identify the segment of the distribution to which each grid cell belongs. Second, we conduct a two-tail trimming of the data by excluding grid cells in the 2.5 percentile bins on both sides of the distribution cumulatively. The main criterion for exclusion is if a grid cell appears in the same bin at least 50 percent of the time (3 out of 6 years).[13] For instance, in the first stage, we exclude grid cells whose malaria-suitability index falls in the 2.5th or 97.5th percentile of the distribution in at least 3 out of 6 years during the study period. In the second stage, we exclude grid cells whose malaria-suitability index falls in the 5th or 95th percentile of the distribution in at least 3 out of 6 years during the study period. We repeat this exercise iteratively until we trim 95 percent of the distribution. In each stage of the trimming exercise, we estimate our baseline model using the student data in the remaining grid cells.

Figure 8 presents the results. The estimated effect of malaria on aggregate test scores persistently remains negative even when we remove 95 percent of the distribution of malaria suitability and the magnitude of the estimated effect is always comparable to the full sample estimation.[14] The coefficients remain statistically significant until we exclude 80 percent of the malaria-suitability distribution. Based on this result we argue that exposure to malaria affects test scores regardless of compliance status.

5.3. Selective Migration

In a broader context, when examining the relationship between malaria and long-term human capital development, migration could be a mechanism through which malaria changes human capital accumula-

13 The results are not sensitive to choosing 2 or 4 out of 6 years.

14 Excluding the observations lying within the remaining 5 percent of the distribution of malaria suitability and repeating the exercise using the remaining observations does not significantly change our main result.

Table 6. Malaria and School Attendance

	Dep. var.: School attendance	
	(1)	(2)
Malaria suitability	−0.0958*	−0.0955*
	(0.0479)	(0.0482)
Female	—	0.0214***
		(0.0042)
Controls	No	Yes
Fixed effects		
YOB	No	Yes
Province	Yes	Yes
Survey round	Yes	Yes
Obs	40,751	38,741
Mean dep. var.	0.6374	0.6390

Source: Regression results are based on students' school attendance data from the Rwanda Household Living Conditions Surveys, and malaria suitability constructed using data from Muñoz Sabater (2019).
Note: Dependent variable is a dummy equal to 1 if the child/student has not missed school during the last 7 days prior to the survey. Controls include gender of household head, gender of the child, rural/urban status, and travel time (minutes) from the house to the nearest primary school. Standard errors in parentheses are clustered by district. *, **, and *** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Nevertheless, the results provide us with some suggestive evidence of how malaria correlates with school attendance, which can be helpful in explaining our baseline results.

Table 6 presents the estimates of the relationship between malaria and school attendance using child-level data from the Rwanda Household Living Conditions Surveys. Our measure of school attendance is an indicator variable equal to 1 if a student has not missed class during the last 7 days preceding the survey during a school season and 0 otherwise. Our main variable of interest is the average standardized malaria-suitability rate in a given district and year. To absorb other correlates of attendance, we control for the time distance between household and school.

The results show a negative association between malaria and the probability of a student attending school. Specifically, a 1-standard-deviation increase in the malaria-suitability index is associated with a 9.5-percentage-point increase in absenteeism (column 2). Relative to the sample mean, this represents a 15 percent increase in absenteeism rate. This effect is substantial and plausibly underlines the effects on test scores observed in earlier sections. In addition, being a female student is associated with higher absenteeism (column 2).

7. Conclusion

In this study, we examine the effects of malaria on human capital accumulation in Africa. To this end, we leverage administrative data on test scores of over a million primary-school students in a high-stakes national exam in Rwanda linked with granular spatial data on malaria suitability between 2012 and 2017. We exploit naturally occurring spatial and temporal variations in the suitability of malaria transmission induced by weather conditions to establish the link between malaria and students' exam performance. The results show that exposure to malaria significantly reduces students' test scores. We also find evidence that malaria widens the gender gap in student performance, as the effect on the test scores of girls is relatively higher than boys. In examining the possible mechanisms, we find absenteeism as a likely channel, as infected children often have to stay home or in a hospital to undergo treatment. Access to health centers and usage of treated bed nets also reduce the effect of malaria.

Finally, we find that increasing malaria suitability in the exam month has adverse effects on students' performance. This suggests that in malaria-endemic countries, moving exam dates to months with the least malaria suitability could play an important role in reducing the impact of malaria on students' performance. We acknowledge a potential limitation in our study. Although the "direct" effects of temperature are accounted for in our estimates, our research design does not explicitly isolate the indirect effect of temperature from the effect of our malaria suitability on test scores. However, the strong correlation between suitability and incidence measures as shown in supplementary online appendix fig. S3.2 suggests that our treatment effects reflect, to a large extent, the impact of malaria on student performance.

Data Availability Statement

Data on students' test scores used in this study were provided by the Rwandan Educational Board, which owns the rights to share the data.

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POLICY RESEARCH WORKING PAPER 4538

License to Sell:

The Effect of Business Registration Reform on Entrepreneurial Activity in Mexico

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Development Research Group
Finance and Private Sector Team
January 2008

Abstract

This paper studies the effect of business registration regulation on economic activity using micro-level data. The identification strategy exploits the fact that a recent business registration reform in Mexico was introduced in different municipalities at different points in time. Using panel data from the Mexican employment survey, I find that the reform increased the number of registered businesses by 5 percent in eligible industries. This increase was due to former wage earners opening businesses. Former unregistered business owners were not more likely to register their business after the reform. Moreover, employment in eligible industries went up by 2.8 percent, and people who were previously unemployed or out of the labor force were more likely to work as wage earners after the reform. Finally, the results imply that the competition from new entrants lowered prices by 0.6 percent and decreased the income of incumbent businesses by 3.2 percent.

This paper—a product of the Finance and Private Sector Team, Development Research Group—is part of a larger effort in the department to understand the impact of business regulation reforms on economic development. Policy Research Working Papers are also posted on the Web at http://econ.worldbank.org. The author may be contacted at mbruhn@worldbank.org.

1 Introduction

Business entry regulation varies widely across the world. Djankov et al (2002), find that the number of procedures for registering a business ranged from 2 in Canada to 21 in the Dominican Republic in 1999. Given these large differences, an important question to study is the effect of entry regulation on economic outcomes. Most work on this question has been based on cross-country studies[1], yet such an approach suffers from endogeneity issues. In particular, cross-country studies cannot rule out that causality goes the other way, such that increases in entry or output lead to reforms. It is also possible that both simpler regulation and higher entry or growth are caused by a third variable. Recent work has also looked at cross-country, cross-industry variation to try to account for endogeneity[2]. However, this approach cannot quantify the overall effect of differences in entry regulation, since all estimates are relative to a benchmark value of the "natural" rates of entry within industries.

This paper uses within country micro-level data to examine the effect of entry regulation on economic outcomes. Specifically, it exploits cross-municipality and cross-time variation in a recent business registration reform in Mexico to measure the effect of this reform on registration, employment, and income, which allows for establishing causality more convincingly than cross-country studies (See Pande and Udry, 2005, for a call to move to research of this type). From a policy perspective, it is also important to study the impact of reforms, since some policy institutions, such as the World Bank, promote simplification of business registration.

The use of micro data also makes it possible to trace out the effects of the reform on the functioning of the product and labor markets. Many economists have argued that barriers to entry harm consumers by raising prices and thwarting employment growth. This paper first examines whether the reform led to the creation of new firms or merely to the registration of existing informal businesses[3]. Having shown that it led to the creation of new businesses, the paper traces out the impact of this increase in competition on consumer prices, incumbents' income, and employment.

The paper starts by building a simple model describing the expected effects on the product and labor markets when the cost of registration drops. In the model, high cost of registration prevents individuals with medium range ability from opening a formal business. Therefore,

[1] For example, Loayza, Oviedo, and Servén (2005) and Djankov, McLiesh, and Ramalho (2006) provide evidence that countries with less regulation grow faster.

[2] Klapper, Laeven, and Rajan (2006), as well as Fisman and Sarria-Allende (2004), show that countries with heavier entry regulation have lower firm entry and lower growth in value added in naturally high-entry industries.

[3] Registration of existing informal businesses may still result in an increase in production and efficiency even if no new businesses are created due to the benefits of formality.

the reform leads to increased entry. Depending on the assumption about returns in the informal sector, the increase in entry comes either from informal business owners registering their business or wage earners opening new businesses. The model also predicts that increased entry leads to a decrease in prices and to a decline in income for incumbent businesses.

The paper then tests the predictions of the model using the Mexican reform. The identification strategy used to estimate the effects of the reform relies on the fact that implementation of the business registration reform in Mexico varied across municipalities and across time. The reform was organized by a federal agency, the COFEMER. This agency had to coordinate with municipality governments on implementing the reform since many business registration procedures are set locally in Mexico. COFEMER's goal was to bring the reform to urban municipalities that had the largest volume of economic activity in Mexico. However, due to staff constraints, COFEMER could not implement the reform in all the priority municipalities at the same time. Moreover, they did not specify a particular pattern of implementation within the set of priority municipalities. A number of checks suggest that the order of implementation was indeed exogenous to the outcomes in this set of municipalities.

The reform reduced registration procedures from 8 on average to less than 3. To give a sense of the magnitude of this reform, the reduction corresponds to going from the 30th percentile in registration procedures to the 2nd percentile, or equivalently it corresponds to going from Peru or Pakistan to New Zealand or Australia. It is important to note that only firms in low-risk to society[4] and unregulated industries were eligible for the reform. These eligible industries encompassed approximately 80 percent of pre-reform firms.

The results show that the reform increased the number of registered businesses by 5 percent in eligible industries, supporting the finding of the cross-country literature that lower regulation leads to more entry. The increase in the number of new businesses came exclusively from former wage earners opening businesses. Informal (non-registered) business owners were not more likely to register their business after the reform. The results also show that employment in eligible industries increased by 2.8 percent after the reform. In particular, people who were previously unemployed or out of the labor force were more likely to work as wage earners after the reform.

By increasing competition, the reform benefitted consumers and hurt incumbent businesses. First, using the Mexican CPI as an outcome variable, I find that the reform decreased the price level by 0.6 percent. The fact that the price decline was concentrated among low-risk industries in the non-tradable goods sector indicates that this was due to competition. Second, the income of incumbent registered businesses declined by 3.2 percent. Some of the evidence also suggests

[4] Low-risk industries are industries that do not present a serious risk to public health, public security, or the environment. An example of a high-risk industry is chemical production.

wage earners. Furthermore, Kaplan, Piedra, and Seira's data does not include information on income. My paper, however, identifies the effect of the reform on the income of different pre-reform occupation groups.

The rest of this paper is organized as follows. Section 2 describes the business registration reform. Section 3 develops a simple occupational choice model which provides the framework for analyzing the effects of the registration reform. Section 4 discusses the identification strategy and Section 5 describes the Mexican employment survey data. Section 6 presents the empirical results. Section 7 concludes.

2 The Mexican Rapid Business Opening System Reform

According to Djankov et al (2002), in 1999, the number of procedures required to register a business in Mexico was 15 and the number of days was 67. Both numbers were above the cross-country average (10 and 48 respectively)[6]. Realizing that Mexico had rather heavy regulations by international comparison, in 2000, the Mexican government created the Federal Commission for Improving Regulation (COFEMER), charged with providing information about the state of regulation across Mexico and implementing possible reforms.

The COFEMER suggested a reform to simplify business registration procedures with the goal of stimulating investment and economic growth. Following this proposal, on March 1, 2002, the Mexican government passed a federal law stating that the number of federal procedures required for starting operation of most businesses should be reduced to a maximum of two procedures that could be administered within 72 hours. These two procedures are obtaining a tax payer number and incorporating the business in case it is a corporation. Once a firm starts operating, it has three months to take care of the other federal requirements that may apply, such as registering workers for medical insurance. This reform applied only to non-governmental firms in industries which do not require special permits or concessions and which do not present a serious risk to public health, public security, or the environment. These eligible "low-risk" industries made up 55 percent of all industries and 80 percent of operating firms, typically micro, small or medium size businesses. Another 10 percent of industries were governmental and 35 percent were classified as "high-risk" or regulated. Appendix A lists examples of low-risk and high-risk/regulated industries as classified by the COFEMER. Examples of low-risk industries are commerce and restaurants. Examples of high-risk or regulated industries are chemical production and transportation (including taxis).

However, simplifying federal regulations was not enough, since there were additional state

[6] The minimum number of procedures (days) was 2 (2) both in Canada and Australia. The maximum number of procedures (days) was 21 in the Dominican Republic (152 in Madagascar). These numbers refer to business registration in the largest city of each country. For Mexico, this is Mexico City.

and municipal procedures required for starting a business. These procedures typically varied from state to state and municipality to municipality. Having simplified federal regulations, the COFEMER then approached state and municipal governments to suggest that they cut down on local regulations and that they implement one-stop-shop centers where entrepreneurs could take care of federal, state, and municipal procedures at the same time. The COFEMER's goal was to create a Rapid Business Opening System (SARE) in Mexico's most populous and economically important urban municipalities in order to quickly reach a large number of people and a large fraction of economic activity with the reform. However, the COFEMER was not able to bring a SARE to all those municipalities at the same time since it had limited resources. There were only four people within the COFEMER working on spreading the reform to local governments. Consequently, the SARE was implemented in different municipalities at different times, staring in May 2002. By September 2006, 103 municipalities had a SARE and another 13 municipalities were in the process of setting up a SARE[7].

The SARE was successful in simplifying local business registration procedures. Table 1 shows summary statistics for business registration procedures before and after the reform for a sample of 32 municipalities from 17 different states. The averages for the number of days, procedures and office visits required to register a business all decreased significantly, falling from 30.1 to 1.4, from 7.9 to 2.7 and from 4.2 to 1, respectively[8]. The standard deviations of all three measures also became much smaller, implying relatively small differences in business registration procedures across municipalities after the reform.

I have fairly detailed administrative data on licenses issued in 2004 from the registration center in one of the municipalities which adopted the reform in 2003, Guadalajara. Looking at these data provides an insight into what types of businesses registered after the reform. Guadalajara reports that a total of 16,631 businesses were created in 2004, corresponding to an investment of US$ 90,997,003 and to 21,170 new jobs created. The average investment was thus US$ 5,471 and the average employment per firm was 1.27. The most frequent business types were video game console rental, computer rental, small grocery stores, clothing stores, home-style food-to-go vendors, and beauty salons.

[7] There are 2454 counties in Mexico, but 94 percent of the population and 98 percent of economic activity are concentrated in 450 counties. These 450 counties include 99 of the 103 counties which had a SARE by September 2006. The SARE counties contain 33 percent of the population and 47 percent of economic activity. The four counties which are not part of the 450 biggest counties, but have a SARE, implemented the reform when other counties in the same state implemented it.

[8] These pre-reform data are different from the data in Djankov et al since the latter are for Mexico City only. The 32 counties for which COFEMER reports data don't include any county from Mexico City since these counties have not implemented reforms.

these municipalities. The map is coded in the following way. The 17 municipalities which adopted the reform early, between May 2002 and March 2004, are marked by circles. The 17 municipalities which adopted the reform late, between April 2004 and December 2004 are marked by triangles. Both early and late adopters are fairly dispersed throughout Mexico.

The fact that adoption of the reform varied across municipalities and across time makes it possible to control for municipality specific and time specific effects. The effect of the reform is thus identified using cross-municipality differences in the reform dummy over time. The identification strategy is valid as long as the changes in outcome variables over time would be similar across municipalities in the absence of the reform. In particular, the identification strategy may be violated if the implementation of the reform followed a specific pattern in terms of municipality characteristics that are related to changes in outcomes. For example, if the municipalities which experienced a high increase in registered businesses adopted the reform first, I might find an effect of the reform even if there was no effect. To gauge whether or not there was such a specific pattern of implementation, I performed a number of checks.

First, I interviewed several staff members at the COFEMER who are in charge of implementing the reform. They informed me that their goal was to bring the reform first to the urban municipalities that have the largest volume of economic activity in Mexico. However, within this set of municipalities they did not specify a particular pattern of implementation. In fact, they mentioned that all municipality governments they approached were very interested in adopting the reform. As discussed in Section 2, COFEMER was not able to implement the reform in all municipalities simultaneously since they did not have enough personnel.

Second, to check whether the 34 municipalities in my sample are indeed comparable, I examine data from the 1994 and 1999 Economic Census. Panel A of Table 3 presents averages for 1999 log GDP per capita, log economic establishments per 1000 capita, log fixed assets per capita, and log investment per capita split-up by early and late adopters[11]. The comparison of early and late adopters in Column 3 of Table 3 shows that they are no statistically significant differences in terms of the number of economic establishments per capita, fixed assets per capita, and investment per capita. Log GDP per capita is higher on average for early adopters. To further examine the pattern of implementation of the reform, I regress each of the three Census variables on the quarter of implementation. The coefficients on quarter of implementation are reported in Column 4 of Table 3. While there is no statistically significant trend for changes in log employment per capita and log investment per capita, log GDP per capita and fixed assets per capita are higher for municipalities that adopted the reform earlier.

[11] The GDP, establishment, fixed assets, and investment data come from the 1994 and 1999 Economic Censuses. To convert these numbers into per capita values I use population data from the 1995 Population Count and the 2000 Demographic Census.

I choose the ENE data as my outcome data since it has five important features for this study. First, it covers almost all municipalities that implemented the reform. Second, the quarterly frequency of the data allows me to exploit the staggered implementation of the reform in the identification strategy. Third, the ENE includes detailed questions about a person's economic activity (including self-employment). Fourth, the ENE captures and distinguishes between formal and informal employment and firms. Fifth, the panel structure of the ENE allows me to investigate who is affected by the reform.

It is important to point out that the way the ENE sample is constructed implies that a municipality-year average is not necessarily representative of the municipality in that year. The sample selection procedure randomly selects households at a geographic unit, the AGEB (Basic Geo-Statistical Area), that is smaller than the municipalities in my sample. All AGEBs within a state are first stratified by socioeconomic characteristics. Within each strata, a certain number of AGEBs is chosen at random. Then, households are chosen at random within the AGEB. This procedure implies that it could happen that only some socioeconomic groups get selected in a given municipality in a given year. However, since the strata are randomly chosen, this remains in expectation a random sample of the households in a municipality, so that the estimate should remain unbiased. Moreover, the survey includes several different strata for most of the municipalities in my sample.

An alternative to using ENE data would have been to use administrative data from business registries. This data is, however, not readily available for Mexico since municipalities do not follow a unified system of recording registered firms. Many records are paper records held only at the local registry and are not publicly available. Moreover, some municipalities updated to an electronic system after the reform, suggesting that the reform also impacted record keeping, which may make pre- and post-reform records incomparable. Finally, unlike the ENE, registry data would not include pre-registration information on firms, such as whether they existed in the informal sector.

Since I am looking at labor market outcomes, I keep only individuals of working age (between the ages of 20 and 65) in my sample. I construct three of my main outcome variables by creating dummy variables for each person in the sample, which indicate whether the person a) is employed, b) is a wage earner, and c) owns a registered business[12]. The later dummies encompass subcategories of the earlier dummies, in the sense that everybody who is employed can either be a wage earner or a business owner (I include the self-employed in the category business owners). Business owners can either be registered or unregistered. However, since all dummies are defined for the whole sample, they denote the fraction of all individuals in my

[12] The Appendix includes a description of how I constructed these dummy variables, paying particular attention to the way business registration is measured in the ENE and explaining some caveats.

variable. Most average changes in outcome variables are not statistically different between early and late adopter municipalities. They are also not significantly correlated with the quarter of implementation, suggesting that the identification strategy is valid. The one exception is the average change in employment. Municipalities that adopted earlier appear to have smaller increases in overall employment. This would bias the analysis against finding an effect of the reform on overall employment.

I also use a number of individual background variables from the ENE as control variables in my regression. These variables include age, gender, marital status, and education dummies. Summary statistics for these background variables and their pre-period changes are listed in the lower panels of Table 4a and 4b. The background variables are very similar across early and late adopter municipalities. The only statistically significant difference in Table 4a is in the fraction of females, where early adopter municipalities have about one percent fewer females.

6 Results

In this section, I put to test the predictions of the model developed above and analyze the effects of the reform on average labor market outcomes. I also break down these effects by pre-reform occupation and examine the impact of the reform on prices.

According to the identification strategy described in Section 4, I obtain the main results by running the following regression with OLS

$$y_{ict} = \alpha + \beta_c + \gamma_t + \delta SARE_{ct} + \pi Z_{ict} + \phi EC_{1999} * t + \varepsilon_{ict},$$

where the subscript i denotes individuals, c denotes municipalities, and t denotes quarters. This regression includes municipality fixed effects, β_c, and quarter fixed effects, γ_t. The variable $SARE_{ct}$ is the reform dummy and, for each municipality, it is equal to one for the quarter in which the reform was implemented and for all following quarters. I run the regression without and with the individual background variables, Z_{ict}. When including individual background variables, I also include variables from the 1999 Economic Census (EC_{1999}) interacted with a linear time trend, t. These variables are log GDP per capita, log number of economic establishments per 1000 capita, log fixed assets per capita, and log investment per capita[14]. Note that I do not include individual dummies or time trends interacted with municipality dummies in my regressions since they take out a lot of the variation in the reform dummy.

The standard errors of the regressions are clustered at the municipality level. As explained in Bertrand, Duflo, and Mullainathan (2004), clustering at the region level helps to prevent the problem that difference-in-difference estimates which use a large number of time periods

[14] The economic variables are converted to per capita levels using 2000 Demographic Census population data.

severely understate uncorrected standard errors if the outcomes are serially correlated.

6.1 Registration

The first prediction of the model above is that the fraction of registered businesses should increase after the reform. Table 5 contains the regressions for the registered business dummy, which denotes the fraction of all people who own a registered business. Both, the specification without controls and the one with individual level and Economic Census controls, show a positive and significant impact of the reform on the number of registered businesses. The increase of 0.32 percentage points is equal to a 3.8 percent increase in registered businesses from the pre-reform level of 8.3 percent.

Columns 3 to 6 of Table 5 show the impact on registered businesses broken down by low-risk and high-risk businesses. Since only the low-risk businesses are eligible for the reform, the increase in registered businesses should only come from low-risk businesses. This is indeed what the results in Columns 3 to 6 confirm. The fraction of low-risk registered businesses increased by 0.37 percentage points (a increase of 5 percent from the pre-reform level of 7.4 percent), while there was no statistically significant change in high-risk registered businesses. Based on the fact that the 34 municipalities in my sample had a total population of 8,285,900 between ages 20 and 65, according to the 2000 Demographic Census, an increase of 0.37 percentage points in low-risk registered businesses corresponds to an increase in 30,678 firms for all 34 municipalities or 902 firms per municipality, on average. The total increase in the number of formal firms is 7.6 times bigger than the increase in formal firms found in Kaplan, Piedra, and Seira (2006), who estimate only 4,029 new formal firms. As explained above, this big difference may be due to the fact that Kaplan, Piedra, and Seira use registration data from the Mexican Social Security Institute (IMSS), which implies that their estimates don't capture firms that have no employees. They do also do not capture firms that obtain an operating license but do not register their workers with the IMSS.

The regression above estimates only the average effect of the reform on registration over the post-treatment period. However, it is also important to check how this effect looks over time. Figures 7 shows a plot of the coefficients of the following regression

$$y_{ict} = \alpha + \beta_c + \gamma_t + \sum \delta_l Quarter_{lc} + \pi Z_{ict} + \varepsilon_{ict},$$

where $Quarter_l$ is a set of dummy variables for lag and lead quarters relative to the time of implementation in a given municipality. For example, $Quarter_{-1}$ is equal to one for the last quarter before the reform was implemented. Since the implementation of the reform was phased in over time and since the data set overs quarters 2000-II to 2004-IV only, I do not observe the outcome data for all lags and leads for all municipalities. For instance, the data set includes information on $Quarter_{+5}$ only for the eight municipalities that implemented the reform in 2003 or earlier. For $Quarter_{+6}$ the data cover only six municipalities, and for $Quarter_{+7}$ and higher only two municipalities. Similarly, for the lags, the data cover all municipalities only for $Quarter_{-8}$ and higher. For this reason, I limit the regression above to observations that fall between $Quarter_{-8}$ and $Quarter_{+6}$. $Quarter_{-8}$ is also the omitted quarter in the regression, to which all other quarters are being compared.

Figure 7 shows the coefficients on the lag and lead dummies for the regression with the low-risk registered business owner dummy as the outcome variable. The two lighter colored lines are the 95 percent confidence bands of the estimates. Between $Quarter_{-8}$ and $Quarter_{-1}$, the estimated differences tend to be negative or close to zero. After implementation of the reform, the differences become positive, reflecting the increase in registration. The only exception to the positive differences after implementation is $Quarter_{+5}$. As mentioned above, the data set covers fewer municipalities for this lead quarter than for previous lead quarters. The decrease in registration may thus be driven by missing data rather than by a actually reversal of the effect of the reform. Note that the coefficient is much greater for $Quarter_{+4}$ than for other post-reform quarters. To make sure that the result in the regression in Column 4 of Table 5 is not only driven by $Quarter_{+4}$, I rerun the regression without $Quarter_{+4}$. The coefficient drops slightly from 0.0037 to 0.0034, and it is significant at the 5 percent level.

To check whether the measured increase of 0.37 percentage points in low-risk registered businesses from Table 5 seems plausible, I use the 2004 administrative data from the business registration center in Guadalajara. First, I calculate how many new businesses the increase of 0.37 percentage points implies for Guadalajara. Given that Guadalajara had a population between the ages of 20 and 65 of 879,019 in 2000, and assuming that the number of additional businesses which came into being due to the reform was equal to the average 0.37 percentage points, this implies that 3,252 new businesses in Guadalajara were created due to the reform. Then, I compare 3,252 to the total number of licenses issued by the registration center in Guadalajara. The center issued 16,613 new licenses in 2004. Given that the reform was implemented in Guadalajara in May 2003, 16,613 is a lower-bound for the total number of businesses created since the reform was implemented. The regression results thus suggest that most of the new licenses issued since the reform in Guadalajara (13361 and up) would have been issued in the absence of the reform as part of the normal turnover in businesses, and that 3,252 of them were issued as a result of the reform.

As mentioned in Section 2, for some municipalities, the COFEMER reports statistics on pre-reform and post-reform registration procedures. These data are available for 27 municipalities mies. Finally, it includes the set of individual background characteristics, Z_{ioct}, and Economic Census variables, EC_{1999}, interacted with a linear time trend. Individuals who are never observed in the pre-reform period are dropped from this analysis, which makes the sample smaller than the one used to determine the main effects above.

Panel A of Table 7 presents the regression results for this analysis. Column 1 shows that past informal business owners are no more likely to register their businesses after the reform. The effect on past wage earners, on the other hand, is positive and statistically significant at the 15.9 percent level. The coefficient may not be more statistically significant, because, as explained in the previous paragraph, the sample is smaller, lowering power.

Panel B of Table 7 shows the effect on past wage earners, split up by whether they were low-risk or high-risk wage earners. The effect is greater for low-risk wage earners, indicating that individuals who started a low-risk business after the reform were mostly working in low-risk sectors before the reform. Overall, the findings in Column 1 of Table 7 are consistent with the assumption of low productivity of the informal sector in the model in Section 3, and they are inconsistent with the high productivity assumption. The results thus suggests that the low-ability (residual sector) view of informal business owners is more appropriate than the De Soto view.

The model with the low productivity assumption also predicts that the wage earners who open a registered business should be the ones with the highest ability levels [17]. To test this, I would need an estimate of their ability. Unfortunately, the dataset does not include any good proxies for wage earners' ability. The only characteristic in the dataset that is specific to wage earners is whether a wage earner has a written contract or not. About 67.4 percent of wage earners had a written contract in the pre-reform period. In order to examine the effect on wage earners in more detail, Table 8 shows the effect broken down by past wage earners who had a written contract and who did not. The effect is four times bigger for wage earners who did not have a written contract and is statistically significant for this group.

Table 9 examines the effect on different groups of non-registered business owners. For non-registered business owners, the data include information on the number of employees in the firm. This variable is a possible proxy for entrepreneurial ability. Although 79.6 percent of informal business owners work alone, some do have employees or partners, and the ones who do may be the higher ability owners. Panel A of Table 9 reports the effect on non-registered business owners who work alone and on non-registered business owners who have either partners or employees in the business. The coefficients are highly statistically insignificant for both

[17] A limitation of the model and its predictions is that they are based on the quite restrictive assumption that the ability relevant for wage work is the same as the ability relevant for opening a business. In future work, it would be interesting to estimate a full Roy model of occupational choice to determine whether the people who open a business are the ones with the highest return to opening a business.

groups, implying that none of these groups is more likely to register after the reform. Panel B of Table 9 illustrates that non-registered business owners who operate on fixed premises (representing only 4 percent of informal business owners), as opposed to conducting their business in mobile stands or going door-to-door, are also not more likely to register their businesses after the reform.

6.2 Employment

The model from Section 3 also predicts that the reform leads to an increase in the number of wage earners. Table 10 reports the regression for the wage earner dummy. Columns 3 and 4 show that the fraction of wage earners increased in low-risk (eligible) industries. This increase in the fraction of wage earners of 0.64 percentage points corresponds to an increase of about 2 percent over the pre-reform fraction of low-risk wage earners. Dividing the increase in the fraction of wage earners (0.64) by the increase in firms (0.37) gives an average firm size of about 1.7 for newly created firms. This number is much smaller than the average size of new firms calculated in Kaplan, Piedra, and Seira (2006), which is 6.3. Since larger firms may be more likely to register their workers with the IMSS, it is perhaps not surprising that Kaplan, Piedra, and Seira capture larger newly created firms on average.

In the model, overall employment in the formal sector also increases since more agents become formal business owners and wage earners. Table 11 examines the impact of the reform on employment. Employment in low-risk industries went up by 1.24 percentage points, which corresponds to a 2.8 percent increase over the pre-reform low-risk employment level.

Figures 8 and 9 show the effect on wage work and employment in low-risk industries broken down by lag and lead quarters relative to the quarter of implementation. These figures were constructed using the same methodology as for Figure 7. The changes in the fraction of wage earners and employment are close to zero until the quarter of implementation. From $Quarter_0$ or $Quarter_1$ on, both variables are significantly and increasingly higher than before.

Similarly to the fraction of registered businesses, the fraction of wage earners and employment in low-risk industries increased more in municipalities where the reduction in registration procedures was greater, as shown in Columns 2 and 3 of Table 6. In a municipality with the average reduction in procedures (5), the fraction of wage earners increased by 1.2 percent and employment increased by 1.1 percent.

In the model, the increase in wage earners comes from informal business owners. Column 2 in Panel A of Table 7 breaks down the effect on the fraction of wage earners in low-risk industries by pre-reform occupation type. The results show that instead of informal business owners, it was mainly individuals who were previously not employed (the unemployed or out of the labor force) who switched to being low-risk wage earners after the reform. The model could where the CPI varies by municipality and month and the subscript m stand for months. The base month for the CPI is June 2002.

Table 12 presents the results for the price regression, for two different time spans. Both regressions include the same set of municipalities, which are only the municipalities that are in the ENE sample. Column 1 of Table 12 corresponds to the time span for which I have ENE data. As predicted by the model above, the coefficient on the reform dummy is negative. In the regression in Column 2, the time period covered goes up to May 2006, which is the last month for which I have the CPI data. The negative coefficient on the reform dummy is similar in magnitude and is statistically significant in this larger sample. The decrease in the consumer price level after the reform confirms the argument that lowering barriers to entry benefits consumers.

The coefficients in Table 12 indicate that the reform decreased the log price level by approximately 0.6 percent. To get a sense of the magnitude of the effect, I compare it to the average inflation rate from 2000 to 2001. This average inflation rate, calculated as the increase in the CPI from each month in 2000 to the same month in 2001, was 5.7 percent. The decrease in the log CPI of approximately 0.6 percent due to the reform is a decrease in inflation of approximately 0.6 percent. A back-of-the-envelope calculation thus suggests that the reform decreased inflation to about 5.1 percent from 5.7 percent.

If the decline in prices is indeed due to the reform, then the decline should be present only in industries that are eligible for the reform (low-risk industries). Moreover, the measured decline in prices should come entirely from the non-tradables sector. This is because the estimation strategy compares prices across municipalities. The prices of non-tradables may decrease locally, but the price of tradables should not be affected by an increase in the supply in some municipalities only[19]. Table 13 breaks down the CPI into tradables and non-tradables, by low-risk and high-risk industries. The results indeed show that prices declined significantly only for non-tradables in low-risk industries.

6.4 Income

The regression results for the last ENE outcome variable, monthly income, are reported in Table 14. This table displays the effect on log income by past occupation group. The model predicts that the reform should decrease the income of incumbent registered business owners. The results in Column 1 indeed show that nominal income decreased for past registered business owners, by 3.2 percent. Given that the reform led to a decrease in the price level, it is also important to examine the impact of the reform on real income, in order to check whether the

[19] Even if these counties were big enough to cause the price of tradables to change, no-arbitrage would imply that there should not be any relative price difference for tradables across counties.

7 Conclusion

This paper uses microeconomic data to analyze the effects of a business registration reform in Mexico on registration, employment, prices, and income. It also traces out the effects of the reform on different pre-reform occupation groups, thereby offering an insight into the channels through which the effects operate.

First, the paper provides evidence that simplifying entry regulation increases the number of registered businesses. After the business registration reform in Mexico, the total number of registered businesses increased by 5 percent in eligible industries. This finding is in line with previous cross-country studies which show that countries with simpler regulation have more business entry. The paper then illustrates that the increase in registered businesses was due to former wage workers opening businesses, and not due to unregistered business owners registering their businesses. This effect is consistent with a model where unregistered business owners are low-ability individuals in a residual sector. The evidence does no support models that are in line with Hernando De Soto's view that informal business owners are medium-range ability individuals who choose a small scale production technology over wage work and who cannot register since registration is too costly and complicated.

The fact that I do not find informal business owners registering their businesses after the reform could also imply that complexity of business registration is not the relevant constraint keeping firms informal. After a firm is registered, it is presumably under more pressure to pay taxes and to comply with labor regulation (even though compliance is far from universal in Mexico, even for registered firms). The associated costs could be so big that changing only registration procedures may not be enough to push informal businesses over the threshold to formality.

This paper also shows that employment in eligible industries increased by 2.8 percent after the reform. In particular, those previously unemployed and out of the labor force were more likely to work as wage earners after the reform as the number of businesses increased. This effect mirrors the cross-country results on output growth, where less complicated regulation is associated with higher growth in output. It also provides evidence that lowering barriers to entry benefits consumers. Another piece of evidence suggesting that consumers benefit from the reform is that prices decreased by 0.6 percent after the reform in Mexico. I attribute this to the fact that new entrants increase total output in a market with a downward sloping demand curve. The price decline was concentrated among low-risk industries in the non-tradable goods sector, which indicates that it was due to competition. The increased competition also decreases income of incumbent firms by 3.2 percent.

Interestingly, this paper does not find an increase in the income of wage earners who opened